

Requirements Traceability Matrix

DCSA Unified Application Access and Integration Prototype (DCSA-UAL)

Field	Value
Product Name	DCSA Unified Application Access and Integration Prototype
Project Acronym	DCSA-UAL
Document Type	Requirements Traceability Matrix (RTM)
Status	Draft v1.0
Date	2026-09-15
Responds To	DCSA Innovation Call #01 — "Unified Application Access and Integration Prototype" (DCSA Innovation Gateway, HS0021-26-CSO-DCSA)
Traced Documents	PRD , FRD , TechArch , UserStories , PERSONAS , JTBD , JOURNEYS , STORY-MAP , UX-Mockup , .planning/PROJECT.md
Baseline Verified	Yes — counts in §2 were recomputed from the source documents for this RTM, not inherited

DEMO — SYNTHETIC DATA ONLY. This matrix traces a demonstration prototype. No real DCSA data, no real PII, no connection to any real government system. Authentication is simulated; no certificate validation occurs. Every identity, case, questionnaire, issue item, and identifier named in this document is fabricated and invalid by construction.

1. Overview

This Requirements Traceability Matrix provides **bidirectional traceability** across the DCSA-UAL specification suite. It answers four questions that a builder, a reviewer, and a test engineer each need to answer from opposite directions: *what implements this requirement*, *what requirement does this code satisfy*, *what verifies it*, and *whose job does it actually serve*.

The suite is unusual in having two parallel traceability spines rather than one, and this document traces both. The **engineering spine** runs PRD feature → FRD requirement → TechArch specification → user story → test case. The **human-centred spine** runs persona → job-to-be-done → journey → story-map release → user story, and it converges on the same story set from the other side. A story that appears on the engineering spine but not the human spine is a story built for the document rather than for a user; a journey stage with no story behind it is a demo step that cannot be walked. Both conditions are reported in §11 rather than smoothed over.

Traceability here is deliberately literal. Every identifier in every cell of every matrix in this document was extracted from the source documents rather than asserted — the FR-ID set from the 182 requirement headings across the twenty-two FRD feature chunks, the US-ID set from the 151 story headings across the twenty epic chunks, the screen inventory from FR-F03-02, the endpoint catalogue from Y1a, the table set from the TechArch hub and spoke DDL, and the human-layer identifiers from PERSONAS, JTBD, JOURNEYS, and STORY-MAP. Where an identifier does not exist, this document says so and classifies the gap; it does not invent one. The single exception is the **TC**- test-case identifier space, which this RTM introduces because the suite did not previously carry one — each TC is derived from a named acceptance criterion in the FRD or the UserStories and from a named suite in FRD F19 / TechArch §14, and no TC asserts behaviour that is not already specified upstream.

The matrix is organised around one non-negotiable claim. The product charter states that if everything else fails, the flagship cross-application workflow must work. §7 therefore isolates that workflow — PRD **F7**, requirements FR-F07a-01...06 and FR-F07b-01...07, journey **JRN-01.01**, job **JTBD-01.1**, stories **US-059...US-069** — into a single end-to-end view, from persona motivation through to the specific test assertions and the specific spoke API calls that prove both systems changed. A reviewer who reads only §7 should be able to satisfy themselves that the central claim is traceable, testable, and demonstrable.

How to read this document. §2 establishes counts and identifier conventions and records the verification of the suite's own reconciliation claims. §3 is the feature-level master matrix. §4 is the PRD-feature requirement detail. §5 is the complete 182-row requirement-level matrix. §6 is the human-centred layer. §7 is the flagship thread. §8 is the test-case catalogue. §9 and §10 trace non-functional requirements and success metrics through specification, implementation, and verification. §11 is the gap analysis. §12 and §13 are change management and sign-off.

2. Requirements Summary

2.1 What is being traced

- 20 PRD features** — F0 ... F19. 15 are P0, 5 are P1, 0 are P2. F7 is P0 and additionally designated the highest-priority feature in the product.
- 6 PRD product principles** — PRIN-01 ... PRIN-06. These carry IDs because seven user stories parent to them instead of to a job-to-be-done.
- 182 FRD functional requirements** — FR-F00-01 ... FR-F19-10, distributed across 22 feature chunks (F07 and F08 each split into an a/b pair) plus six cross-feature chunks (Y0a, Y0b, Y1a, Y1b, Y2, Y3).
- 38 screens** — SCR-01 ... SCR-38, enumerated normatively in FR-F03-02, each with an owning FR, a role set, and a data source.
- 34 hub BFF endpoints** — catalogued in Y1a, every one authorised server-side and every mutating one audited before response.
- 16 Architecture Decision Records** — ADR-001 ... ADR-016, two of which (ADR-012, ADR-013) are recorded deliberate deviations from the FRD.
- 33 hub tables + 25 spoke tables** across 7 isolated schemas (hub, eapp, pvq, iep, pdt, im, cvs).
- 151 user stories** — US-001 ... US-151, across 20 epics whose numbering matches the PRD feature numbering exactly.

- **20 non-functional requirements** — NFR-01 ... NFR-20 .
- **25 success metrics** — SM-01 ... SM-25 .
- **15 risks** — R-01 ... R-15 .
- **4 personas** — PER-01 ... PER-04 . **16 jobs** — JTBD-01.1 ... JTBD-04.4 . **10 journeys** — JRN-01.01 ... JRN-04.03 , 78 journey stages. **3 releases** — R1 , R2 , R3 . **6 demo segments**.
- **176 test cases** — TC-*, introduced by this document across seven test types.

2.2 Verification of the suite's reconciliation claims

The suite was reconciled immediately before this RTM was generated, and two specific claims were handed over with it. Both were re-verified here rather than accepted.

Claim handed over	Verification method	Result
"FRD reports 186 referenced / 186 defined FR-IDs with zero dangling"	Extracted every FR-F{nn}[ab]-{mm} requirement heading across FRD/*.md ; extracted every inline FR-citation across the same corpus; compared the sets	Partially confirmed, count corrected. The correct figure is 182 defined / 182 referenced , not 186. Zero dangling references and zero orphan definitions — the <i>integrity</i> claim holds exactly. The 186 figure appears to be a stale count; no FR-ID is missing and none is duplicated. Recorded as RTM-GAP-11 .
"US-IDs are 151/151"	Extracted every US-\d{3} story heading across UserStories/Epic-*.md ; compared against the Y0-story-index table and the STORY-MAP placement counts	Confirmed. 151 story headings, 151 index rows, 151 placed on the story map, 0 orphans.
Zero dangling FR- references <i>outside</i> the FRD	Extracted FR- citations from TechArch (56 unique), UserStories (168 unique), JOURNEYS (4), STORY-MAP (2), UX-Mockup (64) and compared each set against the 182 defined IDs	Confirmed. Zero dangling references in any downstream document.
Every PRD feature has ≥1 FRD requirement	Grouped the 182 FR-IDs by originating feature	Confirmed. All 20 features covered; minimum 6 requirements (F01, F04, F08b, F15), maximum 12 (F06, F14, F16).
Every user story cites ≥1 FRD requirement	Parsed the **FRD;** trailer on all 151 stories	Confirmed. 151 of 151 stories carry at least one FR citation.

2.3 Identifier conventions

Prefix	Level	Owning document	Example	Count
F{n}	PRD feature	PRD §5	F7	20
PRIN-{nn}	PRD product principle	PRD §3	PRIN-03	6
NFR-{nn}	Non-functional requirement	PRD §6	NFR-13	20
SM-{nn}	Success metric	PRD §7	SM-02	25
R-{nn}	Risk	PRD §8	R-06	15
Q-{nn}	Open question / assumption	PRD §11	Q-01	7
FR-F{nn}[ab]-{mm}	Functional requirement	FRD	FR-F07b-03	182
SCR-{nn}	Screen	FRD FR-F03-02	SCR-20	38
ADR-{nnn}	Architecture decision	TechArch §16	ADR-011	16
US-{nnn}	User story	UserStories	US-063	151
PER-{nn}	Persona	PERSONAS	PER-01	4
JTBD-{nn}. {n}	Job to be done	JTBD	JTBD-01.1	16
JRN-{nn}. {nn}	Journey	JOURNEYS	JRN-01.01	10
R{n}	Release	STORY-MAP	R1	3
TC-{T}--{nnn}	Test case	this document	TC-E-011	176
RTM-GAP-{nn}	Traceability gap	this document	RTM-GAP-01	14

Test-case type codes. TC-U-* unit · TC-I-* integration (API, full hook chain) · TC-AC-* adapter-contract / conformance · TC-E-* end-to-end (browser) · TC-A-* accessibility · TC-RN-* RBAC negative-path · TC-R-* resilience / degradation.

2.4 Requirement distribution by feature

Feature	Priority	FRD requirements	Stories	Screens	Test cases	Owning TechArch chunks
F0 Simulated multi-method MFA authentication	P0	8 (FR-F00-01...08)	8 (US-001...008)	SCR-01...07	8	§8.1, §6.3, §3.4
F1 Unified session and SSO across all spokes	P0	6 (FR-F01-01...06)	6 (US-009...014)	SCR-08 header	7	§8.2, §1.2, §3.4, §7.1
F2 Role- and attribute-based access control	P0	8 (FR-F02-01...08)	11 (US-015...025)	SCR-26, 27, 30	13	§8.3–8.6, §1.2, §3.4
F3 Unified navigation shell and global chrome	P0	8 (FR-F03-01...08)	7 (US-026...032)	SCR-08, 31, 35	7	§12.1, §12.3, §1.5, §8.9
F4 Role-specific personalised dashboard	P0	6 (FR-F04-01...06)	7 (US-033...039)	SCR-09...12	4	§6.5, §1.2, §12.5
F5 Unified work queue	P0	9 (FR-F05-01...09)	10 (US-040...049)	SCR-13	10	§6.5, §1.2, §10.3

F6 Work-item detail and action completion	P0	12 (FR-F06-01...12)	9 (US-050...058)	SCR-14...19	9	\$6.6, \$1.2, \$12.5
F7 Flagship cross-application workflow	P0 — highest	13 (FR-F07a-01...06 , FR-F07b-01...07)	11 (US-059...069)	SCR-13 → 15 → 16 → 20	12	\$11 (all), \$6.7, \$3.7, ADR-010, ADR-011
F8 Adapter framework and application registry	P0	14 (FR-F08a-01...08 , FR-F08b-01...06)	6 (US-070...075)	SCR-22, 23	13	\$5 (all), \$3.5, \$17
F9 Five simulated spoke services	P0	8 (FR-F09-01...08)	5 (US-076...080)	—	6	\$4 (all), \$7, \$1.4
F10 Unified layer API (BFF)	P0	7 (FR-F10-01...07)	5 (US-081...085)	—	8	\$6 (all), \$1.3, ADR-003, ADR-012
F11 Administrator console	P1	7 (FR-F11-01...07)	8 (US-086...093)	SCR-22...27, 29	8	\$6.8, \$10.7, \$3.6
F12 Application registration and onboarding	P1	8 (FR-F12-01...08)	7 (US-094...100)	SCR-28	7	\$17 (all), \$5.4, \$3.5
F13 Immutable audit trail and viewer	P0	8 (FR-F13-01...08)	8 (US-101...108)	SCR-33, 34	8	\$9 (all), \$3.6, \$3.8, ADR-016
F14 USWDS v3 accessible interface	P0	12 (FR-F14-01...12)	10 (US-109...118)	all + SCR-36	12	\$12.3, \$12.4, \$2.3, ADR-008, ADR-014
F15 Notifications, alerts, announcements	P1	6 (FR-F15-01...06)	6 (US-119...124)	SCR-21, 29	6	\$6.8, \$3.7, \$12.5
F16 Health monitoring, resilience, degraded UX	P1	12 (FR-F16-01...12)	8 (US-125...132)	SCR-24, 25, 32, 38	10	\$10 (all), \$1.2, \$3.6, ADR-007
F17 Synthetic seed data corpus	P0	11 (FR-F17-01...11)	5 (US-133...137)	—	9	\$13 (all), \$4.2
F18 Demo operability	P0	9 (FR-F18-01...09)	7 (US-138...144)	SCR-37	9	\$15 (all), \$0.5, \$0.6, ADR-002
F19 Automated test and a11y verification suite	P1	10 (FR-F19-01...10)	7 (US-145...151)	—	10	\$14 (all), \$2.6
Total	—	182	151	38	176	18 chunks

Test-case counts are assigned by owning feature and sum to 176; many individual TCs additionally satisfy requirements in neighbouring features, which is why the per-feature column in \$5 shows cross-feature TC references.

3. Master Traceability Matrix — PRD → FRD → TechArch → UserStories

One row per PRD feature. This is the top-level view; \$5 expands it to requirement granularity.

PRD feature	FRD requirements	TechArch specification	Key tables / endpoints	ADRs	User stories	Journeys
F0 Simulated MFA	FR-F00-01...08	\$8.1 three simulated IdPs; \$6.3 auth & session endpoints; \$3.4 identity DDL	hub.users, hub.user_auth_methods, hub.auth_transactions, hub.user_roles, hub.user_case_assignments · GET /api/auth/methods, POST /api/auth/initiate, POST /api/auth/complete, POST /api/auth/logout	ADR-009	US-001...008	01.01, 01.02, 01.03, 02.01, 03.01
F1 Unified session / SSO	FR-F01-01...06	\$8.2 session & SSO propagation; \$1.2 session service; \$7.1 common spoke contract	hub.sessions, hub.spoke_contexts · GET /api/session, POST /api/session/extend, POST /api/session/active-role · X-UAL-Principal-Assertion (Ed25519)	ADR-009	US-009...014	01.01, 01.02, 02.01, 03.01, 04.01, 04.03
F2 RBAC/ABAC server-side	FR-F02-01...08	\$8.3 PDP; \$8.4 applicant data scoping at the data layer; \$8.5 client hiding is presentation; \$8.6 non-enumeration; \$8.7 zero-trust alignment; \$1.2 PDP + scoped spoke-query wrapper (PEP)	hub.roles, hub.permissions, hub.role_permissions, hub.attribute_rules · pipeline hook 7 · GET /api/entitlements · deriveScope() / enforceScopeOnResult()	ADR-016	US-015...025	01.01, 01.02, 02.01, 03.01, 03.02, 04.01, 04.03
F3 Navigation shell	FR-F03-01...08	\$12.1 routing; \$12.3 USWDS integration; \$1.5 shell layer; \$8.9 the demo banner as an architectural invariant	apps/web/app/(shell)/layout.tsx · SCR-01...38 inventory · GET /api/entitlements, GET /api/search	ADR-014	US-026...032	01.01, 01.02, 03.01, 03.02
F4 Role dashboards	FR-F04-01...06	\$6.5 dashboard endpoint; \$1.2 dashboard composer; \$12.5 uniform state pattern	hub.dashboard_compositions, hub.work_item_counts_cache · GET /api/dashboard · SCR-09...12	ADR-006	01.01, 01.02, 02.02, 03.01, 03.02	
F5 Unified work queue	FR-F05-01...09	\$6.5 work-queue endpoint; \$1.2 work-queue aggregator; \$10.3 partial-failure aggregation	hub.queue_default_views, hub.user_view_preferences · GET /api/work-items with sourceStatus[] · SCR-13	ADR-006, ADR-007	US-040...049	01.01, 01.02, 01.03, 02.01, 02.02, 03.01, 04.01, 04.03
F6 Work-item detail	FR-F06-01...12	\$6.6 work-item endpoints; \$1.2 work-item service; \$12.5 uniform state pattern	GET /api/work-items/{id}, /related, /activity, POST /actions/{actionId} · SCR-14...19	ADR-006	US-050...058	01.01, 01.02, 01.03, 02.01, 02.02, 03.01, 03.02
F7 FLAGSHIP	FR-F07a-01...06 , FR-F07b-01...07	\$11.1–11.8 forward-recovery saga, execution sequence, failure branches, reconciliation record, observed-state confirmation, generic engine, retry worker, post-condition model; \$6.7 the flagship endpoint	hub.orchestration_definitions, hub.orchestration_transactions, hub.orchestration_retry_queue, hub.idempotency_records · POST /api/orchestration/resolve-pvq-issue, POST /api/orchestration/{transactionId}/retry · SCR-20	ADR-010 (retry queue is a table), ADR-011 (forward recovery, not compensating rollback)	US-059...069	01.01 , 02.01
F8 Adapter framework + registry	FR-F08a-01...08 , FR-F08b-01...06	\$5.1 packages/adapter-contract interface; \$5.2 adapter-runtime policy; \$5.3 PVQ implementation sketch; \$5.4 the registry record; \$5.5 capability negotiation; \$5.6 conformance suite; \$3.5 registry DDL; \$17.4 where each surface gets its knowledge	hub.registered_applications, hub.registry_version, hub.retired_application_ids · adapter-rest-json-v1 · seven interface operations · GET /api/registry-version	ADR-001, ADR-007	US-070...075	01.01, 01.02, 01.03, 02.02, 04.01, 04.02, 04.03
F9 Five simulated spokes	FR-F09-01...08	\$4.1 conventions; \$4.2 cross-namespace reference map; \$4.3–4.8 per-spoke DDL; \$4.9 isolation verification checklist; \$7.2 per-spoke surfaces; \$7.3 direct verification; \$1.4 spoke service template	eapp.* (5), pvq.* (3), iep.* (5), pdt.* (5), im.* (5), cvs.* (2) — 7 schemas, 7 roles, zero cross-schema FKs · ports 7101–7106	ADR-004 (schema-per-service, role-per-service), ADR-005	US-076...080	01.01, 01.03, 02.01, 02.02, 03.02, 04.01, 04.02

F10 Unified layer API (BFF)	FR-F10-01...07	\$6.1 how endpoints are declared; \$6.2 common types; \$6.9 cross-cutting API rules; \$1.3 choke point 1 — the eleven-hook edge pipeline	34 endpoints in <code>Y1a · packages/contracts</code> (TypeBox → types, validators, OpenAPI) · boot-time contract check	ADR-003, ADR-012 , ADR-015	US-081...085	01.01, 02.01, 03.02, 04.02
F11 Administrator console	FR-F11-01...07	\$6.8 admin endpoints; \$10.7 integration issue recording; \$3.6 health & issues DDL	<code>hub.application_health</code> , <code>hub.application_health_checks</code> , <code>hub.integration_issues</code> , <code>hub.announcements</code> · GET <code>/api/admin/health</code> , <code>/integration-issues</code> , POST <code>/api/admin/announcements</code> · SCR-22...27, 29	ADR-007	US-086...093	01.03, 04.01, 04.02, 04.03
F12 Registration / onboarding	FR-F12-01...08	\$17.1 the complete onboarding path; \$17.2 live CVS registration script; \$17.3 what was written and what was not; \$17.5 a seventh app with a different backend; \$5.4 registry record; \$3.5 registration drafts	<code>hub.application_registration_drafts</code> , <code>hub.registered_applications</code> · POST <code>/api/admin/applications</code> , <code>/test-connection</code> , PATCH, DELETE · SCR-28 · docs/ONBOARDING-A-NEW-APP.md	ADR-001	US-094...100	01.02, 04.01
F13 Audit trail and viewer	FR-F13-01...08	\$9.1 three layers of enforcement; \$9.2 the interceptor (pipeline hook 9, <code>onSend</code>); \$9.3 record structure and hash chain; \$9.4 coverage; \$9.5 correlated chains; \$9.6 role-scoped visibility; \$9.7 integrity verification; \$3.8 final grants — where immutability actually lives	<code>hub.audit_events</code> , <code>hub.audit_action_types</code> · no UPDATE/DELETE grant for <code>hub_service</code> · GET <code>/api/audit</code> , <code>/chain/{correlationId}</code> , <code>/integrity</code> , <code>/export</code> · SCR-33, 34	ADR-016 (structural enforcement over developer discipline)	US-101...108	01.01, 02.01, 02.02, 03.01, 03.02, 04.01, 04.02, 04.03
F14 USWDS accessible interface	FR-F14-01...12	\$12.3 USWDS integration and token-only theming; \$12.4 accessibility architecture; \$12.6 rendering strategy; \$2.3 frontend stack; \$2.9 Next.js configuration constraints	<code>packages/theme/_uswds-theme.scss</code> (only file with a colour value) · <code>apps/web/components/uswds/</code> · Stylelint no-visual-literals rule	ADR-008 (USWDS wholesale, token-only), ADR-014 (Next.js 15 App Router)	US-109...118	all ten journeys
F15 Notifications & announcements	FR-F15-01...06	\$6.8 notifications endpoints; \$3.7 notifications DDL; \$12.5 uniform state pattern; \$1.2 alert deriver (read-time, never persisted)	<code>hub.alert_read_state</code> , <code>hub.announcements</code> , <code>hub.announcement_dismissals</code> · GET <code>/api/notifications</code> · SCR-21, 29	ADR-006	US-119...124	01.01, 01.02, 02.02, 03.01, 04.02
F16 Health, resilience, degraded UX	FR-F16-01...12	\$10.1 health monitoring; \$10.2 timeout/retry/circuit policy; \$10.3 partial-failure aggregation; \$10.4 the degraded-mode UX contract; \$10.5 failure injection; \$10.6 automatic recovery; \$10.7 integration issue recording; \$10.8 resilience test matrix	<code>hub.application_health</code> , <code>hub.integration_issues</code> · POST <code>/api/admin/failure-injection</code> · <code>/admin/injection</code> per spoke · SCR-24, 25, 32, 38	ADR-007 (hand-written circuit breaker)	US-125...132	01.02, 01.03, 02.02, 04.01, 04.02, 04.03
F17 Synthetic seed corpus	FR-F17-01...11	\$13.1 design properties; \$13.2 pipeline; \$13.3 volume; \$13.4 flagship preconditions; \$13.5 edge-state coverage; \$13.6 personas; \$13.7 obviously synthetic content; \$13.8 validation (startup fails on bad data); \$13.9 reset	<code>packages/seed</code> · <code>UAL_SEED_CONSTANT</code> , <code>UAL_SEED_REFERENCE_DATE</code> · POST <code>/api/admin/reset</code> · docs/SEED-DATA.md	ADR-004	US-133...137	01.01, 01.02, 02.01, 03.01, 03.02, 04.01
F18 Demo operability	FR-F18-01...09	\$15.1 the single command; \$15.2 <code>compose.yaml</code> ; \$15.3 <code>run.sh</code> surface; \$15.4 health endpoints; \$15.5 demo readiness (SCR-37); \$15.6 how an evaluator drives the demo; \$15.7 troubleshooting; \$0.5 topology and port map	11 containers, 1 compose project, 1 network · <code>run.sh</code> · <code>.env</code> as the single port source · GET <code>/api/admin/status</code> · docs/README.md, docs/DEMO-SCRIPTS.md	ADR-002 (Compose, not Kubernetes), ADR-013 (web UI on port 3000)	US-138...144	the scripted demo path (all six segments)
F19 Test & a11y verification	FR-F19-01...10	\$14.1 the pyramid; \$14.2 flagship E2E; \$14.3 RBAC enforcement; \$14.4 adapter conformance; \$14.5 audit coverage and immutability; \$14.6 accessibility verification; \$14.7 link and control integrity crawl; \$14.8 architecture conformance; \$14.9 determinism; \$14.10 reviewer-readable results	<code>flagship.spec.ts</code> , <code>rbac.matrix.spec.ts</code> , <code>rbac.negative.spec.ts</code> , <code>axe.spec.ts</code> , <code>crawl.spec.ts</code> , <code>isolation.spec.ts</code> , <code>resilience.spec.ts</code> , <code>registration.spec.ts</code> , <code>audit.coverage.spec.ts</code> , <code>packages/conformance</code>	ADR-015, ADR-016	US-145...151	03.02, 04.03 <i>(negative-path and resilience assertions)</i>

4. Requirements Detail — PRD features and what they became

Each PRD feature, the requirements derived from it, and the story set that delivers it. Read this section for the *shape* of a feature's traceability; read §5 for the row-by-row detail.

- F0 — Simulated multi-method MFA authentication (P0)**. Eight requirements covering method selection, the CAC/PIV certificate picker, the ECA external-CA path, the generic-MFA two-step with a deterministic demo code, identity-to-role/attribute binding, timeout and re-authentication, logout with full downstream context termination, and — separately and non-negotiably — simulation labelling. `FR-F00-08` exists on its own because the honesty obligation is not a sub-clause of a login screen: it forbids four specific verbs anywhere in user-facing copy, and `TC-A-001` scans for them on every route. Delivered by US-001...008; US-008 is one of the seven stories parented to `PRIN-06` rather than to a job, because no persona's desired outcome is "be told the authentication is fake."
- F1 — Unified session and SSO (P0)**. Six requirements. The load-bearing one is `FR-F01-02`: the hub presents an Ed25519-signed principal assertion to each spoke, spokes verify `aud` and `exp` and reject anything else, and the browser is never in that trust path. `FR-F01-03` is the zero-re-authentication guarantee and is asserted by test rather than by design intent (`TC-E-004`, `TC-E-012`). `FR-F01-06` makes the correlation ID a first-class lifecycle object, which is what later makes `SM-20` provable.
- F2 — RBAC/ABAC enforced server-side (P0)**. Eight requirements and the largest negative-path story set in the suite — six of the eleven stories (US-018...023) are refusals, each with its own dedicated test case. `FR-F02-04` is notable for pushing applicant data isolation down to the data layer rather than leaving it in the handler: the scope becomes SQL predicates in the spoke repository, so a forgotten filter cannot leak. `FR-F02-07` specifies non-enumeration as a measurable property — byte-identical responses and a normalised timing floor — which `TC-RN-001` and `TC-I-011` assert quantitatively.
- F3 — Unified navigation shell (P0)**. Eight requirements, including the normative 38-screen inventory (`FR-F03-02`) that every other document's screen references resolve against, and `FR-F03-03`, the demo banner. The banner requirement is unusually detailed because it carries a real tension: `NFR-13` demands verbatim copy everywhere, `SM-25` demands the applicant's status answer above the fold at 320×568. The FRD resolves it by budget — accessible-DOM copy always present, visible copy permitted to truncate to its bold lede below 640px — and `TC-A-003` evaluates *accessible* text at three viewports so a conformant build cannot fail.
- F4 — Role-specific personalised dashboard (P0)**. Six requirements, one per role plus the composition endpoint and the widget-interaction contract. The four dashboards are traced separately (`FR-F04-02...05`) precisely so `SM-23` — visibly and substantively different, not a relabelled copy — is checkable rather than rhetorical.
- F5 — Unified work queue (P0)**. Nine requirements. `FR-F05-05` is the resilience contract for the queue and is the single requirement most often cited elsewhere in the degraded-experience chain. `FR-F05-02` defines the normalised `WorkItem` shape that makes five heterogeneous sources sortable against each other at all.
- F6 — Work-item detail and action completion (P0)**. Twelve requirements — the largest set alongside F14 and F16 — because this is where work is actually done and where five spoke-specific detail views (`FR-F06-08...11`) each need their own content specification. `FR-F06-05`, the related-items panel, is explicitly the on-ramp to F7.
- F7 — Flagship cross-application workflow (P0, highest)**. Thirteen requirements across two chunks: `F07a` is the user's journey through the workflow, `F07b` is the orchestration behind it. Every one of the eleven stories US-059...069 maps to a distinct step or failure mode, which is deliberate — a regression in any single step is attributable to one story and one test. See §7 for the full thread.
- F8 — Adapter framework and data-driven registry (P0)**. Fourteen requirements, split into the contract (`F08a`) and the registry (`F08b`). `FR-F08b-02` is the no-hard-coding rule and is enforced by a grep-based architecture conformance test, not by convention. This is the feature with the largest requirement-to-story ratio (14 requirements, 6 stories) because much of the contract is machine-verified rather than user-observable — which is why its gap profile in §11 is the way it is.
- F9 — Five simulated spoke services (P0)**. Eight requirements. `FR-F09-01` is the isolation deliverable; `FR-F09-08` is the proof surface that lets an evaluator `curl` a spoke directly and see its own answer. Those two together are what make `SM-03` and `SM-13` demonstrable rather than asserted.

- **F10 — Unified layer API / BFF (P0).** Seven requirements, of which `FR-F10-01` — the eleven-hook edge pipeline — carries the entire zero-trust and audit story structurally. A route that does not declare an action string and request/response schemas fails the boot-time contract check and the process exits non-zero. Starting is not permitted with a route that could bypass the PDP.
- **F11 — Administrator console (P1).** Seven requirements covering inventory, health, the integration-issue log, application detail, manual connection test, announcements, and console-wide conventions. `FR-F11-07` explicitly refuses administrator exemption from authorisation and audit.
- **F12 — Application registration and onboarding (P1).** Eight requirements. This is the extensibility proof, and `FR-F12-06` names the demo sixth application — **CVS**, the Continuous Vetting Service — which ships running but unregistered on port 7106 specifically so registration can be performed live. `Q-06` is closed against it.
- **F13 — Immutable audit trail and viewer (P0).** Eight requirements. `FR-F13-01` makes the audit write part of the mutation path; `FR-F13-03` makes immutability structural — no route, no SQL statement, and no database grant permits modification — and `TC-I-063` asserts all three layers independently.
- **F14 — USWDS v3 accessible interface (P0).** Twelve requirements, treated as functional requirements rather than as quality notes. `FR-F14-12` is the verification gate itself: the build fails on any serious or critical violation. `FR-F14-10` covers the accessibility of error, empty, and loading states — the screens most likely to be built carelessly.
- **F15 — Notifications, alerts, announcements (P1).** Six requirements. `FR-F15-01` keeps alert derivation read-only over spoke data, so an alert never needs to be undone; `FR-F15-05` keeps announcements from ever obscuring the demo banner.
- **F16 — Health monitoring, resilience, degraded-system UX (P1).** Twelve requirements — the full set of behaviours and UX states that keep the prototype usable when a spoke misbehaves, plus `FR-F16-11`, the administrator failure-injection control that makes degradation rehearsable on demand rather than theoretical.
- **F17 — Synthetic seed data corpus (P0).** Eleven requirements. `FR-F17-03` pins the flagship preconditions and `FR-F17-10` fails startup loudly when any of them is missing — a demo starting on broken data is worse than one that refuses to start. `FR-F17-11` is the reset that makes the demo repeatable.
- **F18 — Demo operability (P0).** Nine requirements: one command, per-service control, pre-flight checks, README, the flagship demo script, five secondary scripts, the service status page, clean shutdown, and documented troubleshooting for the three likeliest demo-day failures.
- **F19 — Automated test and accessibility verification suite (P1).** Ten requirements, each naming what is asserted rather than that tests exist. `FR-F19-09` requires the CI artifact to map every FRD requirement to its verifying test **and flag requirements with no test** — which is the machine-readable counterpart of §11 of this document.

5. Requirement-Level Traceability Matrix — all 182 FRD requirements

Every FRD requirement, its explicit TechArch citation where one exists, the stories that cite it, and the test cases that verify it. A `*(feature chunk)*` entry in the TechArch column means the requirement has no *explicit* FR-ID citation in TechArch and is covered by its feature's owning chunk as listed in §2.4 and §3 — TechArch cites 56 of the 182 FR-IDs by ID and covers the remainder structurally by section. Rows marked — **none** in the story column are carried into §11 as `RTM-GAP-01 ... RTM-GAP-04`.

F0 — Simulated multi-method MFA authentication (8 requirements)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F00-01	Authentication method selection (SCR-01)	<i>(feature chunk)</i>	US-001, US-011	TC-E-001
FR-F00-02	CAC/PIV simulated certificate selection (SCR-02)	<i>(feature chunk)</i>	US-002, US-005	TC-E-001, TC-I-001
FR-F00-03	ECA simulated external CA path (SCR-03)	<i>(feature chunk)</i>	US-003	TC-E-002
FR-F00-04	Generic MFA path: identity entry and one-time code (SCR-04, SCR-05)	§00	US-004, US-005	TC-E-003, TC-I-001
FR-F00-05	Synthetic identity to role and attribute binding	<i>(feature chunk)</i>	US-013	TC-I-006, TC-U-001
FR-F00-06	Session timeout, warning, and re-authentication (SCR-06)	<i>(feature chunk)</i>	US-006, US-117	TC-A-014, TC-I-002
FR-F00-07	Logout and full context termination (SCR-07)	<i>(feature chunk)</i>	US-007	TC-I-003
FR-F00-08	Simulation labeling (non-negotiable)	§08	US-008	TC-A-001

F1 — Unified session and SSO across all spokes (6)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F01-01	Single server-side session issuance	§02	— none	TC-U-002
FR-F01-02	Attested principal propagation to spokes	§01, §02, §07	US-010	TC-AC-001
FR-F01-03	Zero re-authentication guarantee	<i>(feature chunk)</i>	US-009, US-011	TC-E-004, TC-E-012, TC-I-004
FR-F01-04	Per-spoke context handles and invalidation	<i>(feature chunk)</i>	— none	TC-I-003
FR-F01-05	Session state surfaced in the UI	<i>(feature chunk)</i>	US-012	TC-I-005
FR-F01-06	Correlation identifier lifecycle	<i>(feature chunk)</i>	US-014	TC-I-007

F2 — Role- and attribute-based access control, server-side (8)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F02-01	The authorization decision function	<i>(feature chunk)</i>	US-010, US-016	TC-I-009, TC-U-002
FR-F02-02	Role permission matrix	<i>(feature chunk)</i>	US-019, US-020, US-023, US-025	TC-I-012, TC-RN-002, TC-RN-003, TC-RN-006
FR-F02-03	Attribute rules layered on role	<i>(feature chunk)</i>	US-016, US-021, US-022	TC-I-009, TC-RN-004, TC-RN-005
FR-F02-04	Applicant data isolation enforced at the data layer	§01, §02, §08	US-016, US-018, US-030	TC-AC-004, TC-I-014, TC-RN-001

FR-F02-05	Action-level authorization and server-computed action lists	§04	US-017, US-019	TC-I-010, TC-RN-002
FR-F02-06	Entitlements endpoint drives navigation and controls	§01, §06	US-015	TC-I-008
FR-F02-07	Denial handling, audit, and non-enumeration (SCR-30)	(feature chunk)	US-018, US-020, US-024	TC-I-011, TC-RN-001, TC-RN-003
FR-F02-08	Role and attribute visibility for administrators	(feature chunk)	US-025	TC-I-012

F3 — Unified navigation shell and global chrome (8)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F03-01	Shell composition and landmark structure (SCR-08)	(feature chunk)	US-026	TC-A-002
FR-F03-02	Screen inventory (every route maps to a real screen)	(feature chunk)	US-028, US-032	TC-E-006, TC-I-013
FR-F03-03	Non-dismissible demo banner	§02	US-008, US-027, US-123, US-150	TC-A-003
FR-F03-04	Role-differentiated primary navigation	(feature chunk)	US-015, US-028	TC-I-008, TC-I-013
FR-F03-05	Session and identity controls in the header	(feature chunk)	US-012, US-013	TC-I-005, TC-I-006
FR-F03-06	Cross-application breadcrumbs	(feature chunk)	US-029, US-061	TC-E-005
FR-F03-07	Global search entry point (SCR-35)	(feature chunk)	US-030	TC-I-014
FR-F03-08	Not-found route (SCR-31) and page templates	(feature chunk)	US-031	TC-I-015

F4 — Role-specific personalised dashboard (6)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F04-01	Dashboard composition endpoint and widget loading model	(feature chunk)	US-033, US-037, US-038	TC-I-016, TC-R-001
FR-F04-02	Investigator dashboard (SCR-09)	(feature chunk)	US-033	TC-E-007
FR-F04-03	Adjudicator dashboard (SCR-10)	(feature chunk)	US-034	TC-E-007
FR-F04-04	Applicant dashboard (SCR-11)	(feature chunk)	US-035	TC-E-007
FR-F04-05	Administrator dashboard (SCR-12)	(feature chunk)	US-036	TC-E-007
FR-F04-06	Widget interaction, linking, and state integrity	(feature chunk)	US-039	TC-E-008

F5 — Unified work queue (9)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F05-01	Work queue screen (SCR-13)	§02	US-040, US-049	TC-I-022
FR-F05-02	Aggregation and normalization across heterogeneous sources	(feature chunk)	US-040, US-070, US-075	TC-I-017, TC-U-003
FR-F05-03	Filtering semantics	§05	US-021, US-042	TC-I-018, TC-RN-004
FR-F05-04	Sorting, pagination, and result counting	(feature chunk)	US-043, US-045	TC-I-019
FR-F05-05	Partial-failure tolerance and degraded queue (the resilience contract)	(feature chunk)	US-048	TC-R-002
FR-F05-06	Search within the queue	(feature chunk)	US-044	TC-I-020
FR-F05-07	Source attribution on every row	(feature chunk)	US-041	TC-U-003
FR-F05-08	Saved default view per role	(feature chunk)	US-046	TC-I-021
FR-F05-09	Queue context preservation	(feature chunk)	US-047	TC-E-009

F6 — Work-item detail and action completion (12)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F06-01	Work-item detail screen (SCR-14)	§02	US-011, US-050, US-058	TC-I-004, TC-I-023, TC-R-003
FR-F06-02	Type-specific detail content	(feature chunk)	US-050	TC-I-023
FR-F06-03	Server-computed action list and rendering	(feature chunk)	US-017, US-051	TC-I-010, TC-I-024
FR-F06-04	Action forms, validation, and submission	(feature chunk)	US-052, US-053, US-057	TC-E-010, TC-I-025, TC-I-026
FR-F06-05	Related items panel (on-ramp to F7)	(feature chunk)	US-055, US-060	TC-I-028
FR-F06-06	Activity history (merged spoke + hub)	(feature chunk)	US-056	TC-I-029
FR-F06-07	Confirmation, failure differentiation, and recovery paths	(feature chunk)	US-054	TC-I-027

FR-F06-08	eApp case view specifics (SCR-15)	<i>(feature chunk)</i>	US-050, US-062	TC-I-023
FR-F06-09	PDT designation view specifics (SCR-17)	<i>(feature chunk)</i>	US-057	TC-E-010
FR-F06-10	IEP applicant status view specifics (SCR-18)	<i>(feature chunk)</i>	US-057	TC-E-010
FR-F06-11	IM case assignment view specifics (SCR-19)	<i>(feature chunk)</i>	US-057, US-058	TC-E-010
FR-F06-12	Return-to-queue navigation	<i>(feature chunk)</i>	US-053	TC-E-009

F7 part a — FLAGSHIP workflow journey (6)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F07a-01	The eApp ↔ PVQ relationship model	<i>(feature chunk)</i>	US-055, US-060, US-080	TC-I-028
FR-F07a-02	Step 1–2: Entry from the queue and case detail (SCR-13 → SCR-15)	<i>(feature chunk)</i>	US-059	TC-E-011
FR-F07a-03	Step 3: Discovery and traversal to the PVQ issue (SCR-15 → SCR-16)	<i>(feature chunk)</i>	US-061, US-062	TC-E-011
FR-F07a-04	Step 4: The resolution form (SCR-16)	<i>(feature chunk)</i>	US-063	TC-E-011, TC-I-030
FR-F07a-05	Step 6: Verification affordance and return path	<i>(feature chunk)</i>	US-067	TC-E-013
FR-F07a-06	Continuity assertions (the acceptance contract)	<i>(feature chunk)</i>	US-009, US-069, US-145	TC-A-005, TC-E-004, TC-E-012

F7 part b — FLAGSHIP workflow orchestration (7)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F07b-01	Orchestration endpoint and execution sequence	<i>(feature chunk)</i>	US-064	TC-E-011, TC-I-030, TC-I-031
FR-F07b-02	Failure of leg 1 (PVQ): clean abort	<i>(feature chunk)</i>	US-064	TC-I-032
FR-F07b-03	Failure of leg 2 (eApp): partial completion and compensation	§04, §07	US-066	TC-R-004
FR-F07b-04	Dual-system confirmation view (SCR-20)	<i>(feature chunk)</i>	US-065	TC-E-011, TC-I-033
FR-F07b-05	Post-condition state model	<i>(feature chunk)</i>	US-067	TC-E-011, TC-E-013
FR-F07b-06	Correlated audit chain for the workflow	<i>(feature chunk)</i>	US-068	TC-I-034
FR-F07b-07	Generalization: orchestration is not special-cased to one workflow	§01	— none	TC-U-004

F8 part a — Adapter contract (8)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F08a-01	Interface operations (the contract)	§00	— none	TC-AC-002
FR-F08a-02	describe() and healthCheck()	§01	US-072	TC-AC-003, TC-I-057
FR-F08a-03	listWorkItems and getWorkItem	§05	— none	TC-AC-004
FR-F08a-04	performAction, getWorkItemSummary, getActivityHistory	§05	— none	TC-AC-005
FR-F08a-05	Timeout, retry, backoff, and circuit-breaking policy	§01, §02, §05, §10	US-073	TC-AC-006, TC-AC-008
FR-F08a-06	Adapter error taxonomy	<i>(feature chunk)</i>	— none	TC-AC-007
FR-F08a-07	Capability negotiation and graceful affordance degradation	§05	US-072	TC-AC-003
FR-F08a-08	Adapter conformance test suite	§02, §05	US-074	TC-AC-002, TC-AC-009, TC-AC-010

F8 part b — Data-driven application registry (6)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F08b-01	Registry record fields	<i>(feature chunk)</i>	US-070	TC-E-015
FR-F08b-02	Registry-derived behavior (the no-hard-coding rule)	§00, §02, §05	US-070	TC-E-015
FR-F08b-03	Runtime enable/disable	<i>(feature chunk)</i>	US-071	TC-E-015
FR-F08b-04	Role visibility and per-application access scoping	<i>(feature chunk)</i>	— none	TC-I-036
FR-F08b-05	Adapter call logging feeding audit and integration issues	§02	US-073	TC-I-037
FR-F08b-06	Registry integrity and startup validation	§00	US-075	TC-I-035

F9 — Five simulated spoke services with isolated namespaces (8)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F09-01	Service isolation (the deliverable)	§00, §04	US-076, US-079	TC-AC-009, TC-I-038, TC-R-005, TC-U-005
FR-F09-02	eApp (Electronic Application)	(feature chunk)	US-078	TC-I-040
FR-F09-03	PVQ (Personnel Vetting Questionnaire)	(feature chunk)	US-078	TC-I-040
FR-F09-04	IEP (Individual Engagement Portal)	(feature chunk)	US-078	TC-I-040
FR-F09-05	PDT (Position Designation Tool)	(feature chunk)	US-078	TC-I-040
FR-F09-06	IM (Investigation Management)	§04	US-078	TC-I-040
FR-F09-07	Common spoke service requirements	§01, §04	— none	TC-AC-001, TC-AC-011, TC-I-031
FR-F09-08	Independent queryability (the proof surface)	§00	US-067, US-077	TC-E-013, TC-I-039

F10 — Unified layer API / BFF (7)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F10-01	Request pipeline (the choke point)	§01, §02, §16	US-010, US-081	TC-I-041, TC-RN-007, TC-RN-008
FR-F10-02	Endpoint groups and their screens	(feature chunk)	— none	TC-I-045, TC-RN-009
FR-F10-03	Consistent error contract	(feature chunk)	US-082	TC-AC-007, TC-I-011, TC-I-042
FR-F10-04	Pagination, filtering, and sorting conventions	(feature chunk)	— none	TC-I-019, TC-I-046
FR-F10-05	Mandatory audit on mutation	(feature chunk)	US-083	TC-I-043
FR-F10-06	API documentation as a deliverable	§00, §02, §06	US-084	TC-I-044
FR-F10-07	Rate limiting and abuse resistance (demo-grade)	§02	— none	TC-I-047

F11 — Administrator console (7)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F11-01	Connected applications inventory (SCR-22)	(feature chunk)	US-086	TC-I-048
FR-F11-02	System health view (SCR-24)	(feature chunk)	US-087	TC-I-049
FR-F11-03	Integration issues log (SCR-25)	(feature chunk)	US-066, US-088	TC-I-050, TC-R-004
FR-F11-04	Application detail view (SCR-23)	(feature chunk)	US-071, US-075, US-089	TC-I-051
FR-F11-05	Manual connection test	(feature chunk)	US-090	TC-I-052
FR-F11-06	System announcements management (SCR-29)	(feature chunk)	US-091	TC-I-053
FR-F11-07	Policy visibility and console-wide conventions	(feature chunk)	US-020, US-023, US-025, US-092, US-093	TC-A-006, TC-I-054, TC-RN-003, TC-RN-006

F12 — Application registration and onboarding (8)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F12-01	Multi-step registration form (SCR-28)	(feature chunk)	US-094	TC-E-016
FR-F12-02	Field validation rules and messages	(feature chunk)	US-095	TC-I-055
FR-F12-03	Live connection test (step 3)	(feature chunk)	US-096	TC-I-056
FR-F12-04	Capability auto-discovery (step 4)	§05	US-097	TC-I-057
FR-F12-05	Post-registration propagation	(feature chunk)	US-098	TC-E-017
FR-F12-06	The demo sixth application (CVS)	(feature chunk)	US-099	TC-E-016
FR-F12-07	Edit and de-register flows	§03	US-100	TC-I-058
FR-F12-08	Onboarding documentation	(feature chunk)	— none	TC-I-059

F13 — Immutable audit trail and viewer (8)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F13-01	Mandatory audit write on every mutation	(feature chunk)	US-053, US-083, US-101	TC-I-026, TC-I-043, TC-I-060

FR-F13-02	Record schema and coverage	<i>(feature chunk)</i>	US-002, US-093, US-102, US-103	TC-I-054, TC-I-061, TC-I-062
FR-F13-03	Append-only storage and integrity	§02, §03, §04, §06	US-023, US-104	TC-I-063, TC-RN-006
FR-F13-04	Correlated cross-system chains	<i>(feature chunk)</i>	US-014, US-068, US-107	TC-I-007, TC-I-034
FR-F13-05	Audit viewer (SCR-33)	<i>(feature chunk)</i>	US-105	TC-I-064
FR-F13-06	Record detail and chain view (SCR-34)	<i>(feature chunk)</i>	US-106	TC-I-065
FR-F13-07	Role-scoped audit visibility	<i>(feature chunk)</i>	US-107	TC-RN-010
FR-F13-08	Export	<i>(feature chunk)</i>	US-108	TC-I-066

F14 — USWDS v3 accessible interface, Section 508 / WCAG 2.1 AA (12)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F14-01	USWDS component adoption and token-only theming	§02	US-118	TC-A-016, TC-U-006
FR-F14-02	Keyboard operability	<i>(feature chunk)</i>	US-069, US-109	TC-A-005, TC-A-007, TC-A-009
FR-F14-03	Accessible forms	§02, §16	US-052, US-063, US-113	TC-A-010, TC-I-025
FR-F14-04	Accessible data tables	<i>(feature chunk)</i>	US-045, US-092, US-110	TC-A-004, TC-A-006, TC-A-008, TC-I-019
FR-F14-05	Accessible navigation and landmarks	<i>(feature chunk)</i>	US-026, US-111, US-112	TC-A-002, TC-A-007
FR-F14-06	Color contrast and non-color meaning	<i>(feature chunk)</i>	US-111, US-114	TC-A-009, TC-A-011
FR-F14-07	Screen-reader semantics and live regions	<i>(feature chunk)</i>	US-115	TC-A-008, TC-A-012
FR-F14-08	Motion, zoom, and responsive behavior	<i>(feature chunk)</i>	US-035, US-116	TC-A-013
FR-F14-09	Timing and session accessibility	<i>(feature chunk)</i>	US-006, US-117	TC-A-014
FR-F14-10	Error, empty, and loading state accessibility	<i>(feature chunk)</i>	— none	TC-A-015
FR-F14-11	Accessibility statement (SCR-36)	<i>(feature chunk)</i>	US-118	TC-A-016
FR-F14-12	Accessibility verification gate	§02	US-118, US-149	TC-A-017

F15 — Notifications, alerts, and system announcements (6)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F15-01	Derived alert computation	<i>(feature chunk)</i>	US-119, US-124	TC-I-067, TC-R-006
FR-F15-02	Alert presentation: dashboard, header indicator, list (SCR-21)	<i>(feature chunk)</i>	US-120, US-124	TC-I-068
FR-F15-03	Read/unread state	<i>(feature chunk)</i>	US-121	TC-I-069
FR-F15-04	System announcements rendering	<i>(feature chunk)</i>	US-091, US-122	TC-I-053, TC-I-070
FR-F15-05	Distinction from the demo banner	<i>(feature chunk)</i>	US-027, US-123	TC-A-003, TC-A-018
FR-F15-06	Notifications list screen behavior (SCR-21)	<i>(feature chunk)</i>	US-120	TC-I-068

F16 — Health monitoring, resilience, degraded-system UX (12)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F16-01	Health monitor service	<i>(feature chunk)</i>	US-125	TC-I-071
FR-F16-02	Health summary endpoint	<i>(feature chunk)</i>	US-125	TC-I-049, TC-I-071
FR-F16-03	Health state definitions	§10	US-087, US-125	TC-AC-008, TC-I-071
FR-F16-04	Pre-emptive action disabling	<i>(feature chunk)</i>	US-051, US-126	TC-I-024, TC-R-007
FR-F16-05	Degraded-system warning presentation	<i>(feature chunk)</i>	US-038, US-048, US-127	TC-R-001, TC-R-002, TC-R-003, TC-R-008
FR-F16-06	Loading states	§02	US-037, US-128	TC-I-016, TC-R-009
FR-F16-07	Empty states	§10	US-049, US-124, US-129	TC-I-022, TC-R-006, TC-R-010, TC-R-014
FR-F16-08	Error states	<i>(feature chunk)</i>	US-054, US-058, US-130	TC-I-015, TC-I-027, TC-R-003, TC-R-011
FR-F16-09	Integration issue recording	<i>(feature chunk)</i>	US-088	TC-I-037, TC-I-050
FR-F16-10	Global error boundary (SCR-32)	<i>(feature chunk)</i>	US-130	TC-R-011
FR-F16-11	Failure injection controls (SCR-38)	§01, §04	US-131	TC-I-072
FR-F16-12	Automatic recovery	<i>(feature chunk)</i>	US-038, US-132	TC-R-001, TC-R-012

F17 — Synthetic seed data corpus (11)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F17-01	Seed volume and distribution	(feature chunk)	US-133	TC-I-073
FR-F17-02	Personas and role/attribute coverage	§13	— none	TC-I-074
FR-F17-03	The flagship workflow's seeded preconditions	(feature chunk)	US-137	TC-I-075
FR-F17-04	Referential coherence across isolated namespaces	§04	US-080	TC-U-005, TC-U-007
FR-F17-05	Deterministic seeding	(feature chunk)	US-136	TC-U-008
FR-F17-06	Edge-state coverage	(feature chunk)	US-022, US-134	TC-I-076, TC-RN-005
FR-F17-07	Per-persona screen coverage matrix	(feature chunk)	US-133	TC-I-073
FR-F17-08	Obviously synthetic content	§01, §04	US-135	TC-U-009
FR-F17-09	Seed data documentation	§00	US-135	TC-U-009
FR-F17-10	Seed validation	(feature chunk)	US-137	TC-I-075, TC-I-077
FR-F17-11	Reset to baseline	§09	US-099, US-136	TC-E-014, TC-E-018

F18 — Demo operability (9)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F18-01	Single startup command	§00, §02	US-138, US-139	TC-E-019
FR-F18-02	Per-service control	§00	US-079, US-143	TC-I-078, TC-R-005
FR-F18-03	Pre-flight checks, seeding, and reset	(feature chunk)	US-139	TC-I-079
FR-F18-04	README	§00, §02	US-144	TC-E-020
FR-F18-05	Flagship demo script (F7)	§00	US-059, US-077, US-140	TC-E-021, TC-I-039
FR-F18-06	Secondary demo scripts	§00	US-141	TC-E-022
FR-F18-07	Service status page (SCR-37)	(feature chunk)	US-142	TC-I-080
FR-F18-08	Clean shutdown	(feature chunk)	US-144	TC-I-081
FR-F18-09	Documented troubleshooting	(feature chunk)	US-144	TC-I-082

F19 — Automated test and accessibility verification suite (10)

FRD requirement	Title	TechArch cite	User stories	Test cases
FR-F19-01	Flagship workflow end-to-end test	(feature chunk)	US-145	TC-E-014, TC-E-023
FR-F19-02	RBAC enforcement tests	§08	US-018, US-019, US-085, US-146	TC-RN-001, TC-RN-007, TC-RN-008, TC-RN-009, TC-RN-011
FR-F19-03	Adapter conformance and behavior tests	§00, §01, §03, §04	US-074, US-147	TC-AC-010, TC-AC-012
FR-F19-04	Resilience tests	§06, §10	US-147	TC-R-013, TC-R-014, TC-R-015
FR-F19-05	Audit coverage and immutability tests	(feature chunk)	US-148	TC-I-083
FR-F19-06	Automated accessibility scan	§02	US-149	TC-A-017, TC-A-019
FR-F19-07	Keyboard navigation smoke test	§02	US-149	TC-A-005, TC-A-007, TC-A-020
FR-F19-08	Link and control integrity crawl	(feature chunk)	US-032, US-150	TC-A-003, TC-E-006, TC-E-024
FR-F19-09	Reviewer-readable results summary	(feature chunk)	US-085, US-151	TC-I-084
FR-F19-10	Test environment and determinism	§02	US-151	TC-I-085, TC-U-008

6. Human-Centred Traceability Layer — PERSONAS → JTBD → JOURNEYS → STORY-MAP → UserStories

The suite carries a full human-centred layer, and a builder needs it as much as the engineering spine: it is what tells you *why* a requirement exists and *which demo segment breaks* if you cut a story. This section traces it end to end and converges it on the same 151 stories.

6.1 Personas → Jobs

Persona	Role	Jobs held	Primary job	Release at which they can do meaningful work	Demo segments
PER-01 Marcus Vale	Investigator	JTBD-01.1 , JTBD-01.2 , JTBD-01.3 , JTBD-01.4	JTBD-01.1 — clear a blocking issue in one sitting (<i>the flagship</i>)	R1 — fully	1 (<i>primary</i>), 2a, 3b
PER-02 Dana Okonkwo	Adjudicator	JTBD-02.1 , JTBD-02.2 , JTBD-02.3 , JTBD-02.4	JTBD-02.2 — render a determination defensible years later	R2 — fully; none in R1 (<i>deliberate</i>)	4, 2c
PER-03 Renée Ashford	Applicant	JTBD-03.1 , JTBD-03.2 , JTBD-03.3 , JTBD-03.4	JTBD-03.1 — find out where I actually stand	R2 — fully; none in R1 (<i>deliberate</i>)	5a, 5b
PER-04 Priya Raghunathan	Administrator	JTBD-04.1 , JTBD-04.2 , JTBD-04.3 , JTBD-04.4	JTBD-04.1 — onboard the next application as configuration	R1 partial (<i>verifier only</i>) → R2 partial (<i>auditor</i>) → R3 fully (<i>operator</i>)	3a, 6, 2b

Seeded identity binding is asserted at seed time by FR-F17-02 AC-3 and FR-F17-10 , and verified by TC-I-074 : PER-01 → Marcus Vale, PER-02 → Dana Okonkwo, PER-03 → Renée Ashford (SUBJ-00622), PER-04 → Priya Raghunathan. The supporting cast (Harlan T. Boyce, Ingrid L. Vasterling, Sofia K. Mendelbaum, Theodore Q. Lansbury / SUBJ-00418 , Bartholomew N. Quigley) are deliberately **not** design personas and carry no PER-ID .

6.2 Jobs → Features → Journeys → Stories

Story sets are taken from the STORY-MAP JTBD coverage table, which is the authoritative job→story mapping in the suite.

JTBD	Persona	Priority	PRD features	Journeys	Stories carrying the job	NaC derived	First release
JTBD-01.1 (<i>flagship</i>)	PER-01	P0	F7 , F1, F2, F3, F6, F9, F10, F13, F14	JRN-01.01	US-001, US-002, US-009, US-016, US-021, US-022, US-055, US-059...US-069 , US-077, US-109	10	R1
JTBD-01.2	PER-01	P0	F5, F4, F15, F3, F2, F17	JRN-01.02, JRN-01.01	US-033, US-039, US-040...US-047, US-119	5	R1
JTBD-01.3	PER-01	P0	F13, F6, F10, F2, F1	JRN-01.01, JRN-01.02	US-053, US-056, US-068, US-083, US-101, US-106	4	R1
JTBD-01.4	PER-01	P1	F16, F5, F4, F14, F9, F11	JRN-01.03, JRN-04.03	US-037, US-048, US-058, US-073, US-115, US-126, US-127, US-128, US-130, US-132	6	R2 (<i>US-130</i>) → R3
JTBD-02.1	PER-02	P0	F6, F5, F7, F1, F3, F14, F17	JRN-02.01	US-011, US-030, US-041, US-044, US-050, US-055, US-056, US-114	4	R2
JTBD-02.2	PER-02	P0	F2, F6, F13, F10, F14	JRN-02.01	US-013, US-017, US-019, US-051, US-057, US-101, US-102, US-104, US-105, US-146	5	R2
JTBD-02.3	PER-02	P1	F4, F5, F15, F6, F14	JRN-02.02	US-034, US-039, US-042, US-043, US-057, US-120, US-121	3	R2
JTBD-02.4	PER-02	P1	F16, F6, F5, F14, F9	JRN-02.02	US-038, US-049, US-058, US-124, US-126, US-127, US-129	3	R2 (<i>US-049</i>) → R3
JTBD-03.1	PER-03	P0	F4, F5, F1, F3, F14, F17	JRN-03.01, JRN-03.02	US-003, US-004, US-015, US-028, US-035, US-116	4	R2
JTBD-03.2	PER-03	P0	F5, F6, F4, F0, F14	JRN-03.01	US-006, US-052, US-053, US-057, US-113, US-117	4	R2
JTBD-03.3	PER-03	P1	F15, F4, F6, F14, F3	JRN-03.01	US-115, US-121, US-122, US-123	3	R2
JTBD-03.4	PER-03	P1	F2, F13, F6, F10, F19	JRN-03.02	US-005, US-007, US-015, US-018, US-020, US-024, US-103, US-105, US-107	5	R1 (<i>partial</i>) → R2
JTBD-04.1 (<i>extensibility</i>)	PER-04	P1	F12 , F8, F11, F13, F14, F9, F18	JRN-04.01, JRN-01.02	US-070, US-071, US-074, US-094...US-100 , US-147	5	R1 (<i>partial</i>) → R3
JTBD-04.2	PER-04	P1	F11 , F16 , F8, F14, F9, F10	JRN-04.02, JRN-04.03	US-036, US-086, US-087, US-090, US-125, US-131, US-132, US-149, US-151	4	R1 (<i>partial</i>) → R3
JTBD-04.3	PER-04	P1	F11, F13, F16, F10, F14, F8	JRN-04.02	US-025, US-082, US-084, US-088, US-089, US-091, US-103, US-105, US-139, US-144	5	R2 (<i>partial</i>) → R3
JTBD-04.4	PER-04	P1	F13, F11, F16, F2, F15, F19, F18	JRN-04.02, JRN-04.03, JRN-03.02	US-014, US-023, US-081, US-093, US-102, US-103, US-104, US-108, US-131, US-141, US-143, US-148	4	R1 (<i>partial</i>) → R3

Job coverage: 16 of 16 addressed. 74 needs-and-constraints derived. The flagship job carries 10 — the heaviest derivation in the suite, which is correct: it has the most stages, the most failure modes, and the most at stake in a live evaluation.

6.3 Journeys → Stories → Releases → Demo segments

Journey	Persona	Stages	Primary jobs	Stage-covering stories	Completed in	Demo segment
JRN-01.01 Clear the blocking issue (<i>flagship</i>)	PER-01	11	JTBD-01.1 (<i>primary</i>), JTBD-01.2 (<i>entry</i>), JTBD-01.3 (<i>record</i>)	US-059...US-069 (11/11)	R1	1 — primary, ~5 min
JRN-01.02 Monday-morning triage	PER-01	7	JTBD-01.2 , JTBD-01.3	US-033, US-039, US-040...US-047, US-119 (<i>stages 1–6</i>); US-098 (<i>stage 7</i>)	R3 (<i>6/7 at R1</i>)	3b
JRN-01.03 Keep working while IM is down	PER-01	7	JTBD-01.4	US-048, US-126, US-127, US-128, US-132 + US-037, US-058, US-073, US-115, US-130	R3	2a
JRN-02.01 Assemble and determine	PER-02	9	JTBD-02.1 , JTBD-02.2	US-011, US-013, US-017, US-019, US-030, US-041, US-044, US-050, US-051, US-055, US-056, US-057, US-114	R2 (<i>9/9</i>)	4
JRN-02.02 Aging queue, missing data	PER-02	7	JTBD-02.3 , JTBD-02.4	US-034, US-038, US-039, US-042, US-043, US-049, US-120, US-121, US-124, US-126, US-127; US-129 (<i>stages 5–6</i>)	R3 (<i>5/7 at R2</i>)	2c
JRN-03.01 Where I stand, what I owe	PER-03	8	JTBD-03.1 , JTBD-03.2 , JTBD-03.3	US-003, US-004, US-006, US-015, US-028, US-035, US-052, US-053, US-057, US-113, US-115, US-116, US-117, US-121, US-122, US-123	R2 (<i>8/8</i>)	5a
JRN-03.02 Proof and boundary	PER-03	6	JTBD-03.4	US-005, US-007, US-015, US-018, US-020, US-024 , US-103, US-105, US-107	R2 (<i>6/6</i>)	5b
JRN-04.01 Register the sixth application	PER-04	9	JTBD-04.1	US-094...US-100 + US-070, US-071, US-074, US-147	R3 (<i>9/9</i>)	3a
JRN-04.02 Triage an integration failure	PER-04	8	JTBD-04.2 , JTBD-04.3 , JTBD-04.4	US-086...US-093 + US-025, US-082, US-084, US-103, US-105	R3 (<i>8/8</i>)	6
JRN-04.03 Rehearse degradation	PER-04	6	JTBD-04.2 , JTBD-04.4	US-131, US-143, US-125, US-132 + US-141, US-127	R3 (<i>6/6</i>)	2b

Journey coverage: 10 of 10 journeys and 78 of 78 stages have at least one covering story.

Traceability-direction note, carried to RTM-GAP-05 . JOURNEYS-DCSA-UAL .md contains **zero US- references** across all thirteen of its chunks (verified). Journey→story traceability therefore exists only in STORY-MAP-DCSA-UAL .md , and the story sets in the table above are reconstructed from the STORY-MAP JTBD coverage table and the per-release "closed by" attributions. The reconstruction is sound and the coverage claim is verifiable, but a JOURNEYS revision should carry story IDs on its stage tables so the link is stated rather than derived.

6.4 Story-map releases → stories → demonstrable journeys

Release	Theme	Stories	Backbone coverage	Journeys complete	Demo segments runnable
R1	Walking skeleton — the flagship journey, end to end	67	Authenticate 8 · Orient 7 · Find work 14 · Work an item 17 · Resolve across systems 8 · Verify & Audit 10 · Administer & Extend 3	JRN-01.01 (11/11)	Segment 1 (the thesis)
R2	Role breadth and trust	34 (36 rows less 2 split-acceptance carry-overs)	Authenticate 7 · Orient 8 · Find work 5 (incl. US-078 IEP/PDT and US-133 PER-02...04, both continuing from R1) · Work an item 3 · Resolve across systems 9 · Verify & Audit 4	+ JRN-02.01, JRN-03.01, JRN-03.02	Segments 1, 4, 5a, 5b (role- and resource-level zero trust)
R3	Extensibility and operability	50	Orient 4 · Find work 2 · Work an item 2 · Verify & Audit 10 · Administer & Extend 32	+ JRN-01.02, JRN-01.03, JRN-02.02, JRN-04.01, JRN-04.02, JRN-04.03 — all ten	All six segments
Total	—	151	7 of 7 backbone steps populated	10 of 10	6 of 6

Three stories (US-061, US-067, US-107) appear in a second activity row marked (shared) because they carry two distinct user-observable claims at the same stage; three (US-078, US-107, US-133) have acceptance legitimately split across two releases. Neither is double-counted in the totals. Seven stories (US-008, US-027, US-032, US-118, US-123, US-135, US-150) have **no JTBD ancestor** and are parented to PRD product principles — US-008, US-027, US-118, US-123, US-135 → PRIN-06 (honest demo); US-032, US-150 → PRIN-03 (every button works).

6.5 Principles → Stories

Principle	Statement	Stories parented to it	Requirements	Test cases
PRIN-01	Continuity over navigation	(delivered through JTBD-01.1)	FR-F01-03, FR-F03-06, FR-F07a-03, FR-F07a-06	TC-E-005, TC-E-012
PRIN-02	Server is the authority	(delivered through F2/F10)	FR-F02-01, FR-F02-06, FR-F10-01	TC-I-041, TC-RN-007, TC-RN-011
PRIN-03	Every button works	US-032, US-150	FR-F03-02, FR-F19-08	TC-E-006, TC-E-024
PRIN-04	Degrade visibly, never blankly	(delivered through F16)	FR-F16-05, FR-F16-08, FR-F16-10	TC-R-008, TC-R-011, TC-R-013
PRIN-05	Configuration over code	(delivered through F8/F12)	FR-F08b-02, FR-F12-01, FR-F12-05	TC-E-015, TC-E-016, TC-E-017
PRIN-06	Honest demo	US-008, US-027, US-118, US-123, US-135	FR-F00-08, FR-F03-03, FR-F14-11, FR-F15-05, FR-F17-08	TC-A-001, TC-A-003, TC-A-016, TC-A-018, TC-U-009

7. FLAGSHIP Traceability Thread — the product's central claim, end to end

The charter is unambiguous: "If everything else fails, that single unified workflow must work." This section exists so a reviewer can verify that claim's traceability in one view rather than by assembling it from six documents.

7.1 The claim

An investigator, signed in **once**, opens an eApp case from the unified work queue, discovers a related PVQ issue raised against a specific questionnaire answer, resolves that issue **without leaving the unified experience**, and observes **both** eApp and PVQ reflect the change — with no second login, no second application, no re-entry of context, and no manual correlation.

7.2 The thread

Layer	Identifier	Content
Charter	.planning/PROJECT.md "Core Value"	"A user signs in once and completes a cross-application workflow end-to-end without ever logging into — or navigating to — a second system."
Persona	PER-01 Marcus Vale	Field/desk background investigator, 38–45 active cases across IM, eApp, PVQ, PDT; measured on on-time closure and on the defensibility of what he recorded
Job	JTBD-01.1 (P0)	"When a case is blocked by an issue raised against a questionnaire answer, I want to clear it in one sitting, so the investigation keeps moving."
Journey	JRN-01.01 — 11 stages, Demo Segment 1, target under three minutes manually	Sign in → Orient → Enter the work → Open the case → Read the answer → Discover the relationship → Traverse → Resolve → See both systems answer → Verify on the case → Leave the record
PRD feature	F7 (P0 — highest priority in the product)	Eleven named capabilities from entry through orchestration, consistency handling, dual-system confirmation, verification affordance, audit chain, continuity assertions, and the scripted demo path
Screens	SCR-13 → SCR-15 → SCR-16 → SCR-20, then SCR-15 re-read → SCR-34	Work queue → eApp case view → PVQ issue detail & resolution → dual-system confirmation → case re-read → audit chain view
Endpoints	GET /api/work-items → GET /api/work-items/EAPP:CASE-A-1042 → /related → GET /api/work-items/PVQ:ISS-2207 → POST /api/orchestration/resolve-pvq-issue → GET /api/audit/chain/{correlationId}; retry via POST /api/orchestration/{transactionId}/retry	The orchestration endpoint is the only one whose ActionDescriptor.targetSystems has length > 1

Seeded preconditions	FR-F17-03 , JRN-01.01 P1–P8	hub: INVESTIGATOR bound to PER-01, CAC/PIV, org=DCSA-FIELD-OPS-EAST , tier=T5 , region=REGION-NE , holds the assignment · eapp: CASE-A-1042 , SUBJ-00418 , UNDER_REVIEW , outstandingIssueCount=1 , due in 4 days, ≥2 employer entries in SECTION_13A · pvq: ISS-2207 , OPEN , answerLocus=SECTION_13A.employer[0] .endDate , raised 3 days ago, answerSnapshot populated · pdt + im: designations/assignments on the same case so the panel shows three relationship types · hub: default queue surfaces the case on page 1 unfiltered, ALERT-NEW-PVQ-ISSUE fires, ~120 pre-seeded audit events, plus a spare second open issue for mid-session re-runs
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7.3 Requirements → stories → tests, step by step

#	Journey stage	FRD requirement	User story	Verifying test cases
—	Relationship model (<i>underpins 6–10</i>)	FR-F07a-01 — the eApp ↔ PVQ relationship model	US-055, US-060, US-080	TC-I-028, TC-U-005, TC-U-007
1–3	Sign in, orient, enter the work	FR-F00-01 , FR-F00-02 , FR-F04-02 , FR-F15-01 , FR-F05-01	US-001, US-002, US-033, US-119, US-040	TC-E-001, TC-E-007, TC-I-016, TC-I-067, TC-I-017
4–5	Open the case, read the flagged answer	FR-F07a-02 — entry from the queue and case detail (SCR-13 → SCR-15)	US-059	TC-E-011, TC-E-012, TC-I-023
6	Discover the relationship	FR-F07a-01 , FR-F06-05 — related-items panel resolved live through the PVQ adapter	US-060	TC-E-011, TC-I-028
7	Traverse (<i>highest accessibility risk in the product</i>)	FR-F07a-03 — discovery and traversal (SCR-15 → SCR-16); FR-F03-06 breadcrumbs	US-061 , US-062	TC-E-011, TC-E-012, TC-E-005, TC-A-005
8	Resolve — disposition and narrative	FR-F07a-04 — the resolution form (SCR-16); FR-F14-03 accessible forms	US-063	TC-E-011, TC-I-025, TC-I-030, TC-A-010
8	Orchestrate both legs	FR-F07b-01 execution sequence; FR-F07b-02 leg-1 clean abort	US-064	TC-E-011, TC-I-030, TC-I-031, TC-I-032
9	See both systems answer	FR-F07b-04 dual-system confirmation (SCR-20); FR-F07b-05 post-condition state model	US-065	TC-E-011, TC-I-033
9'	Partial completion — the honest half	FR-F07b-03 leg-2 failure, partial completion and forward recovery; FR-F11-03 integration issue	US-066	TC-R-004 , TC-I-050
10	Verify independently	FR-F07a-05 verification affordance and return path; FR-F09-08 independent queryability	US-067	TC-E-013 , TC-I-039, TC-E-009
11	Leave the record	FR-F07b-06 correlated audit chain; FR-F13-04 cross-system chains	US-068	TC-I-034, TC-I-007, TC-I-065
all	Keyboard-only completion	FR-F14-02 keyboard operability; FR-F07a-06 continuity assertions	US-069	TC-A-005 , TC-A-007, TC-A-009
all	Continuity contract	FR-F07a-06 — the acceptance contract, ten asserted properties	US-009, US-069, US-145	TC-E-012 , TC-E-004
—	Generic orchestration (<i>not special-cased</i>)	FR-F07b-07 — the engine names no spoke	— no story (<i>RTM-GAP-02</i>)	TC-U-004
—	Demo script and repeatability	FR-F18-05 flagship demo script steps 1–13; FR-F17-11 reset; FR-F19-01 E2E gate	US-140, US-136, US-145	TC-E-021, TC-E-014, TC-E-018, TC-E-023

7.4 Architecture that carries it

Concern	TechArch specification	ADR
Distributed write	§11.2 execution sequence — reconciliation row written before the first spoke call; PVQ leg first (authoritative outcome), then eApp parent-state	—
Failure strategy	§11.1, §11.3 — forward recovery, never compensating rollback : a recorded investigative disposition is not fabricated away	ADR-011
Retry durability	§11.7 retry worker draining <code>hub.orchestration_retry_queue</code> on a backoff schedule	ADR-010 — the retry queue is a database table, not in-process state
Confirmation integrity	§11.5 — the confirmation view shows observed state, not asserted state : a fresh re-read per spoke, and a failed re-read renders "couldn't confirm" rather than being omitted	—
Genericity	§11.6 — the engine is driven by <code>hub.orchestration_definitions</code> and contains no application identifier; enforced by architecture conformance test	ADR-016
Idempotency	<code>hub.idempotency_records</code> at the hub, <code><ns>.idempotency_records</code> at each spoke, 24-hour retention, 409 IDEMPOTENCY_KEY_REUSED on same key + different payload	—
Authorisation of both legs	§8.3 PDP evaluates <code>ISSUE.RESOLVE</code> on <code>PVQ:ISS-2207</code> and <code>CASE.UPDATE_ISSUE_STATE</code> on <code>EAPP:CASE-A-1042</code> before anything is written	ADR-016
Audit	§9.2 interceptor at pipeline hook 9 (<code>onSend</code> , mutations), §9.5 correlated chains	ADR-016

7.5 Success metrics riding on the flagship

Metric	Target	Verified by	Observed where
SM-01 Flagship workflow completion	End to end in a single session, under three minutes manually	TC-E-011, TC-E-023	Live demo Segment 1 + <code>flagship.spec.ts</code>
SM-02 Re-authentications	Exactly zero — one <code>AUTH_SUCCESS</code> record for the session	TC-E-012, TC-E-004	<code>flagship.spec.ts:authEventCount</code> + audit log shown live
SM-03 Dual-system change verified independently	Both spoke APIs return updated state with the hub out of the path	TC-E-013 , TC-I-039	<code>curl http://localhost:7102/issues/ISS-2207</code> and <code>:7101/cases/CASE-A-1042</code> , live
SM-04 Context loss	Zero identifiers re-entered at any step	TC-E-012	<code>identifiersTypedByTest()</code> equals the narrative only
SM-08 Keyboard-only completion	Flagship completable keyboard-only	TC-A-005	Recorded manual keyboard pass + keyboard smoke suite
SM-20 Flagship audit chain	Whole action retrievable as one correlated chain	TC-I-034	Audit viewer SCR-34, live
SM-22 Demo repeatability	Three consecutive identical runs with a reset between	TC-E-014, TC-E-018	Pre-demo rehearsal

7.6 Risks attached to this thread

Risk	Mitigation traced to
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R-02	Flagship feels stitched together	FR-F07a-03 in-shell traversal, FR-F03-06 breadcrumb carry-over, FR-F07a-06 asserted continuity — TC-E-012, TC-E-005
R-03	Scope breadth starves the flagship	STORY-MAP R1 scope test (<i>is this slice needed to watch JRN-01.01 complete?</i>); US-145 ships in R1 so R3's breadth cannot silently break R1's depth
R-06	Dual-system update partially fails during the demo	FR-F07b-03 explicit partial-completion state, no "success" copy, retry path, forward recovery — TC-R-004 asserts the absence of the word "success" literally
R-08	Demo environment fails on demo day	FR-F18-01, FR-F18-03, FR-F18-09, FR-F17-11 — TC-E-019, TC-I-079, TC-I-082, TC-E-018

8. Test Case Coverage Matrix

176 test cases across seven types. Every test case is derived from a named acceptance criterion in the FRD or in a user story, and is assigned to a suite named in `FR-F19-01..10` or `TechArch §14`. Coverage properties, verified by construction:

- **All 182 FRD requirements have at least one verifying test case.**
- **All 151 user stories have at least one verifying test case.**
- **All 20 PRD features have test cases in at least two different test types.**
- No test case asserts behaviour that is not already specified upstream; where a TC exists to close a story gap (for example `TC-U-004` for `FR-F07b-07`), it is flagged in §11.

Type	Count	Tier / tooling	Runtime budget	What it protects
Unit	9	Vitest 2.1 (~400 assertions)	< 30 s	PDP decisions, normalisation, circuit state machine, hash chain, scope derivation, date offsetting, architecture invariants
Integration (API)	85	Vitest + <code>fastify.inject()</code> (~250)	< 90 s	The complete hook chain per endpoint: session → CSRF → reserved fields → validation → PDP → handler → audit
Adapter-contract	12	@ual/conformance (10 × 6 adapters)	< 60 s	The adapter contract, per adapter, standalone-runnable and run live during registration
End-to-end	24	Playwright 1.49 (~25 specs)	< 6 min	Flagship, RBAC, degraded, registration, keyboard-only, crawl
Accessibility	20	@axe-core/playwright — every route × every role × 5 states	< 5 min	WCAG 2.1 AA / Section 508 gate
RBAC negative-path	11	Generated from the role matrix + <code>rbac.negative.spec.ts</code>	(<i>in integration budget</i>)	Zero-trust enforcement, non-enumeration, denial auditing
Resilience / degradation	15	<code>resilience.spec.ts</code> with failure injection	(<i>in E2E budget</i>)	Graceful degradation, partial-failure honesty, automatic recovery

8.1 Unit — 9 cases

TC ID	F	Assertion	FRD source	Stories	Verified by
TC-U-001	F0	Each synthetic identity resolves to its declared role set and attribute bundle (organization, clearanceTier, assignedRegion, caseAssignments, subjectRef); a multi-role identity resolves to all roles held.	FR-F00-05	US-013	Vitest unit tier
TC-U-002	F1	Principal is rebuilt from <code>hub.sessions</code> on every request; no role, identity, or scope is read from the cookie or request body.	FR-F01-01, FR-F02-01	US-010	Vitest unit tier
TC-U-003	F5	Normalization: every spoke-native status maps to exactly one <code>statusCategory</code> ; priority maps onto the three-tier scale; <code>workItemId</code> is <code>{sourceSystem}:{nativeId}</code> ; <code>overdue</code> is computed hub-side.	FR-F05-02, FR-F05-07	US-041	Vitest unit tier
TC-U-004	F7	Architecture conformance: the orchestration engine source contains no reference to <code>PVQ</code> , <code>EAPP</code> , or any other application identifier — the saga is generic and registry-driven.	FR-F07b-07	—	Architecture conformance (TechArch §14.8)
TC-U-005	F9	Cross-namespace relationships are opaque references resolved by the hub; no join, no foreign key, and no shared schema crosses a namespace boundary.	FR-F09-01, FR-F17-04	US-080	Vitest unit + <code>information_schema</code> introspection
TC-U-006	F14	Stylelint finds zero hard-coded colour, font, or spacing literals across all SCSS and component styles; <code>packages/theme/_uswds-theme.scss</code> is the only file containing a colour value.	FR-F14-01	US-118	Stylelint gate (NFR-03)
TC-U-007	F17	Referential coherence: every <code>subjectRef</code> , <code>parentCaseRef</code> , and <code>answerLocus</code> in the seed resolves against the owning namespace, including the one intentional seeded orphan.	FR-F17-04	US-080	Seed validator unit tests
TC-U-008	F17	Determinism: the same <code>UAL_SEED_CONSTANT</code> produces byte-identical seed output; all relative dates derive from <code>UAL_SEED_REFERENCE_DATE</code> , never from wall-clock <code>now</code> .	FR-F17-05, FR-F19-10	US-136	Vitest unit tier
TC-U-009	F17	Every spoke response body carries <code>_synthetic: true</code> and every record a <code>syntheticMarker</code> ; name, address, and identifier generators are invalid-by-construction.	FR-F17-08, FR-F17-09	US-135	Vitest unit tier

8.2 Integration (API — full hook chain via `fastify.inject()`) — 85 cases

TC ID	F	Assertion	FRD source	Stories	Verified by
TC-I-001	F0	A failed sign-in returns a byte-identical non-disclosing error for a valid and an invented identity; nothing in the response reveals whether the account exists.	FR-F00-02, FR-F00-04	US-005	Vitest + <code>fastify.inject()</code>
TC-I-002	F0	Idle timeout fires the SCR-06 warning with a live countdown; <code>POST /api/session/extend</code> extends in place; entered form data is not discarded.	FR-F00-06	US-006	Vitest + <code>fastify.inject()</code>
TC-I-003	F0	<code>POST /api/auth/logout</code> terminates the hub session and invalidates every <code>hub.spoke_contexts</code> row; a subsequent adapter call with the old context is rejected.	FR-F00-07, FR-F01-04	US-007	Vitest + <code>fastify.inject()</code>
TC-I-004	F1	A deep link to a work-item route survives one sign-in and lands on the requested item via <code>returnTo</code> , with no second credential prompt.	FR-F01-03, FR-F06-01	US-011	Vitest + <code>fastify.inject()</code>
TC-I-005	F1	<code>GET /api/session</code> returns identity, <code>activeRole</code> , and remaining session time; the header renders them and the values match the server.	FR-F01-05, FR-F03-05	US-012	Vitest + <code>fastify.inject()</code>
TC-I-006	F1	<code>POST /api/session/active-role</code> switches the active role within the held role set without re-authentication, is audited, and re-computes entitlements.	FR-F00-05, FR-F03-05	US-013	Vitest + <code>fastify.inject()</code>

TC-I-007	F1	One correlation ID generated at the hub edge appears on every adapter call header, every audit row, every integration-issue row, and the error envelope for that action.	FR-F01-06 , FR-F13-04	US-014	Vitest + fastify.inject()
TC-I-008	F2	GET /api/entitlements returns the server-computed navigation and permission set; the rendered nav and controls are a strict subset of it.	FR-F02-06 , FR-F03-04	US-015	Vitest + fastify.inject()
TC-I-009	F2	Resource-level authorization: fetching work item X verifies this principal's entitlement to X specifically, not merely to the route.	FR-F02-01 , FR-F02-03	US-016	Vitest + fastify.inject()
TC-I-010	F2	The same work item returns different ActionDescriptor[] sets for an investigator and an adjudicator; each action is re-authorized at execution time.	FR-F02-05 , FR-F06-03	US-017	Vitest + fastify.inject()
TC-I-011	F2	Every denial returns the same envelope shape, code, and copy; response bodies for a forbidden ID and a fabricated ID differ only in correlationId , and timings are normalized to the 120 ms floor.	FR-F02-07 , FR-F10-03	US-024	rbac.negative.spec.ts
TC-I-012	F2	SCR-26/27 expose each identity's roles, attributes, and recent activity to administrators; the role matrix and attribute rules are inspectable rather than inferred.	FR-F02-08 , FR-F02-02	US-025	Vitest + fastify.inject()
TC-I-013	F3	Primary navigation is generated per role from entitlements; the four nav sets are distinct and every item maps to a row in the SCR-01...38 inventory.	FR-F03-04 , FR-F03-02	US-028	Vitest + fastify.inject()
TC-I-014	F3	GET /api/search is scoped server-side to work items the principal may see; a search for another subject's identifier returns zero rows rather than a denial.	FR-F03-07 , FR-F02-04	US-030	Vitest + fastify.inject()
TC-I-015	F3	A fabricated URL renders SCR-31 inside the shell with working exits to dashboard and queue — never a framework 404 page and never a stack trace.	FR-F03-08 , FR-F16-08	US-031	Vitest + fastify.inject()
TC-I-016	F4	GET /api/dashboard returns role-appropriate widget data with per-source status; a slow source degrades one widget and the remaining widgets render.	FR-F04-01 , FR-F16-06	US-033, US-037	Vitest + fastify.inject()
TC-I-017	F5	GET /api/work-items fans out concurrently over the registry rows visible to the active role and returns correctly attributed items from at least four of five spokes.	FR-F05-02	US-040	Vitest + fastify.inject()
TC-I-018	F5	Filtering by source system, type, status, priority, assignee, and due-date range; active filters render as removable chips with a working clear-all.	FR-F05-03	US-042	Vitest + fastify.inject()
TC-I-019	F5	Sorting on due date, priority, status, source, and last activity; pagination with an announced result count; sortable headers announce sort state.	FR-F05-04 , FR-F14-04 , FR-F10-04	US-043, US-045	Vitest + fastify.inject()
TC-I-020	F5	Search across title, subject reference, and identifier is applied server-side within the principal's scope.	FR-F05-06	US-044	Vitest + fastify.inject()
TC-I-021	F5	Each role opens the queue on its saved default view (investigator: assigned to me, due date ascending).	FR-F05-08	US-046	Vitest + fastify.inject()
TC-I-022	F5	A genuine zero-result queue renders the designed empty state with guidance; an unreachable source renders the degraded state instead — the two are never the same screen.	FR-F05-01 , FR-F16-07	US-049, US-129	Vitest + fastify.inject()
TC-I-023	F6	GET /api/work-items/{id} returns full detail with source attribution and subject context; eApp case detail renders its questionnaire sections.	FR-F06-01 , FR-F06-02 , FR-F06-08	US-050	Vitest + fastify.inject()
TC-I-024	F6	Only server-computed actions render; a disabled action carries a plain-language disabledReason ; an action targeting a DOWN system is pre-emptively disabled.	FR-F06-03 , FR-F16-04	US-051, US-126	Vitest + fastify.inject()
TC-I-025	F6	Action forms validate against the same TypeBox schema client- and server-side; invalid submission returns field errors and the typed narrative is preserved verbatim.	FR-F06-04 , FR-F14-03	US-052	Vitest + fastify.inject()
TC-I-026	F6	The confirmation names exactly what changed and in which system, and the state shown is the spoke's confirmed write — never an optimistic value.	FR-F06-04 , FR-F13-01	US-053	Vitest + fastify.inject()
TC-I-027	F6	Authorization denial, validation failure, and source unavailability produce three distinct, recoverable presentations, each carrying a correlation ID.	FR-F06-07 , FR-F16-08	US-054	Vitest + fastify.inject()
TC-I-028	F6	GET /api/work-items/{id}/related resolves RelatedRefs live through the target adapters; the relationship label comes from the spoke's record, not the UI.	FR-F06-05 , FR-F07a-01	US-055, US-060	Vitest + fastify.inject()
TC-I-029	F6	GET /api/work-items/{id}/activity merges spoke-native history with hub audit records into one chronology showing actor, action, timestamp, and origin system.	FR-F06-06	US-056	Vitest + fastify.inject()
TC-I-030	F7	The orchestration endpoint authorizes both legs — ISSUE.RESOLVE on PVQ:ISS-2207 and CASE.UPDATE_ISSUE_STATE on EAPP:CASE-A-1042 — before any write; a client-supplied parent case reference that does not match the issue's real parent is rejected.	FR-F07b-01 , FR-F07a-04	US-063, US-064	flagship.spec.ts + integration tier
TC-I-031	F7	Replaying the orchestration with the same X-UAL-Idempotency-Key returns the stored outcome and executes neither leg a second time, at hub and at spoke.	FR-F07b-01 , FR-F09-07	US-064	Vitest + fastify.inject()
TC-I-032	F7	Leg-1 (PVQ) failure aborts cleanly: no eApp call is attempted, the response states unambiguously that nothing changed, and a reconciliation row exists.	FR-F07b-02	US-064	Vitest + fastify.inject()
TC-I-033	F7	SCR-20 shows a per-system results table whose values are a fresh re-read from each owning spoke; a failed re-read renders a 'couldn't confirm' row with a working Check-again control rather than being omitted.	FR-F07b-04	US-065	flagship.spec.ts
TC-I-034	F7	GET /api/audit/chain/{correlationId} returns ≥5 records spanning case read, related-item resolution, traversal, issue read, both writes, and completion, with a summary line and before/after summaries.	FR-F07b-06 , FR-F13-04	US-068	flagship.spec.ts:chain

TC-I-035	F8	Registry startup validation rejects a duplicate identifier, a retired identifier, a malformed endpoint, and an unresolvable <code>adapterType</code> , and the hub exits non-zero with an actionable message.	FR-F08b-06	US-075	Vitest + real Postgres
TC-I-036	F8	Per-application role visibility scoping: an application visible only to ADMINISTRATOR never appears in an investigator's navigation, queue fan-out, or search.	FR-F08b-04	—	Vitest + <code>fastify.inject()</code>
TC-I-037	F8	Every adapter call emits a structured log carrying the correlation ID, and a failure emits exactly one <code>hub.integration_issues</code> row linked to the corresponding audit record.	FR-F08b-05 , FR-F16-09	US-073, US-088	Vitest + real Postgres
TC-I-038	F9	Isolation: 42 cross-schema read attempts (every service role × every foreign schema) raise <code>42501</code> , and <code>information_schema</code> shows zero cross-schema foreign keys.	FR-F09-01	US-076	<code>isolation.spec.ts</code>
TC-I-039	F9	Each spoke's published port answers directly with a short-lived operator token: <code>GET {pvq}/issues/ISS-2207</code> and <code>GET {eapp}/cases/CASE-A-1042</code> return the spoke's own state with the hub out of the path.	FR-F09-08 , FR-F18-05	US-077	<code>isolation.spec.ts</code> + <code>flagship.spec.ts:spokeGet</code>
TC-I-040	F9	Per-spoke domain behaviour: eApp case state transitions, PVQ issue disposition, IEP notices/tasks, PDT tier derivation, IM assignment and leads each behave as specified.	FR-F09-02 , FR-F09-03 , FR-F09-04 , FR-F09-05 , FR-F09-06	US-078	Vitest + real Postgres, per service
TC-I-041	F10	The eleven ordered edge hooks run for every route in every child plugin; a route declared without an <code>action</code> string or request/response schema fails the boot-time contract check and the process exits non-zero.	FR-F10-01	US-081	Boot-time contract check + integration tier
TC-I-042	F10	Every non-2xx response from every endpoint matches the error envelope: machine code, human message, detail, correlation ID, field errors, retryability — and contains no stack trace, hostname, port, or SQL fragment.	FR-F10-03	US-082	Vitest + <code>fastify.inject()</code> , all routes
TC-I-043	F10	Every mutating route in the route table produces exactly one audit record of its declared action type, written before the success response is observable.	FR-F10-05 , FR-F13-01	US-083, US-148	<code>audit.coverage.spec.ts</code>
TC-I-044	F10	The generated OpenAPI document matches the route declarations field for field; hand-edited documentation cannot drift because none exists.	FR-F10-06	US-084	Architecture conformance (TechArch §14.8)
TC-I-045	F10	Every endpoint group in <code>Y1a</code> names the screen(s) it serves and every screen in the inventory names the endpoint(s) it consumes; the mapping is complete in both directions.	FR-F10-02	—	Contract cross-check (generated)
TC-I-046	F10	Pagination, filtering, and sorting parameter conventions are identical across every collection endpoint, including the announced total and the partial-source-failure report.	FR-F10-04	—	Vitest + <code>fastify.inject()</code>
TC-I-047	F10	Demo-grade rate limiting bounds repeated authentication and orchestration attempts without impeding the scripted demo path.	FR-F10-07	—	Vitest + <code>fastify.inject()</code>
TC-I-048	F11	SCR-22 lists every registered application with identifier, adapter type, endpoint, work-item types, actions, enabled state, and registration date, read directly from the registry.	FR-F11-01	US-086	Vitest + <code>fastify.inject()</code>
TC-I-049	F11	SCR-24 reports per-application status, last successful check, latency, and recent check history from <code>hub.application_health</code> / <code>application_health_checks</code> .	FR-F11-02 , FR-F16-02	US-087	Vitest + <code>fastify.inject()</code>
TC-I-050	F11	An induced adapter failure produces a correctly attributed SCR-25 entry within one health-check interval, filterable, and linked to its audit record by correlation ID.	FR-F11-03 , FR-F16-09	US-088	<code>resilience.spec.ts</code>
TC-I-051	F11	SCR-23 shows one application's full configuration, health history, and recent errors in a single view.	FR-F11-04	US-089	Vitest + <code>fastify.inject()</code>
TC-I-052	F11	<code>POST /api/admin/applications/{id}/probe</code> performs a live health check on demand and reports the real result, including failure.	FR-F11-05	US-090	Vitest + <code>fastify.inject()</code>
TC-I-053	F11	Announcement create/edit/expire with severity, target roles, and effective/expiry dates; an announcement targeted at one role appears on that role's dashboard and no other.	FR-F11-06 , FR-F15-04	US-091	Vitest + <code>fastify.inject()</code>
TC-I-054	F11	Every administrator console action is authorized by the PDP and writes an audit record; the administrator is not exempt from either.	FR-F11-07 , FR-F13-02	US-093	<code>audit.coverage.spec.ts</code>
TC-I-055	F12	Registration validation: duplicate identifier, retired identifier, unreachable endpoint, and malformed policy values each produce the specified accessible inline error and error-summary entry.	FR-F12-02	US-095	<code>registration.spec.ts</code>
TC-I-056	F12	<code>POST /api/admin/applications/test-connection</code> gates submission: an unreachable or non-conformant candidate cannot be registered.	FR-F12-03	US-096	<code>registration.spec.ts</code>
TC-I-057	F12	Capability auto-discovery pre-populates work-item types and actions from the candidate's own <code>describe()</code> and presents them for confirmation rather than transcription.	FR-F12-04 , FR-F08a-02	US-097	<code>registration.spec.ts</code>
TC-I-058	F12	Edit and de-register require confirmation, write audit records naming the administrator and the configuration, and retire the identifier against reuse.	FR-F12-07	US-100	<code>registration.spec.ts</code>
TC-I-059	F12	<code>docs/ONBOARDING-A-NEW-APP.md</code> states everything a new application must implement to be registerable, and the stated list matches what the conformance suite actually enforces.	FR-F12-08	—	Documentation conformance check
TC-I-060	F13	Exactly one audit record per successful mutation; failing the audit store causes the mutation to report failure rather than proceed silently.	FR-F13-01	US-101	<code>audit.coverage.spec.ts</code>
TC-I-061	F13	The record captures actor identity and the roles/attributes held at the time of the action , plus target system, resource, outcome, correlation ID, and a before/after summary.	FR-F13-02	US-102	<code>audit.coverage.spec.ts</code>
TC-I-062	F13	Coverage beyond mutations: authentication success and failure, authorization denials, registration changes, and adapter failures are all present in the trail.	FR-F13-02	US-103	<code>audit.coverage.spec.ts</code>

TC-I-063	F13	Append-only: route enumeration finds no PUT/PATCH/DELETE under <code>/api/audit</code> ; static analysis finds no UPDATE/DELETE against <code>audit_events</code> ; <code>UPDATE hub.audit_events as hub_service</code> raises <code>42501</code> ; the hash chain verifies and fails at the exact altered sequence number, with no gaps across a full run.	FR-F13-03	US-104	<code>audit.coverage.spec.ts</code> + DB grant tier
TC-I-064	F13	SCR-33 filters by actor, role, action type, target system, resource, outcome, and date range, sorted and paginated.	FR-F13-05	US-105	Vitest + <code>fastify.inject()</code>
TC-I-065	F13	SCR-34 opens one record in full and follows it into the chain sharing its correlation ID.	FR-F13-06	US-106	Vitest + <code>fastify.inject()</code>
TC-I-066	F13	<code>GET /api/audit/export</code> returns exactly the records shown on screen for the active filter and carries the synthetic-data notice.	FR-F13-08	US-108	Vitest + <code>fastify.inject()</code>
TC-I-067	F15	Alerts are derived read-only at read time from aggregated spoke data — overdue, newly assigned, newly raised PVQ issue on an owned case, blocked, approaching due date — and no derived alert is persisted.	FR-F15-01	US-119	Vitest + <code>fastify.inject()</code>
TC-I-068	F15	The header indicator count reconciles with the notifications list, and every alert resolves to the real populated work item that produced it.	FR-F15-02, FR-F15-06	US-120	Vitest + <code>fastify.inject()</code> + <code>crawl.spec.ts</code>
TC-I-069	F15	Read/unread state persists per user and per alert; marking an alert read does not remove it from the list.	FR-F15-03	US-121	Vitest + <code>fastify.inject()</code>
TC-I-070	F15	Announcements render as USWDS site alerts, are dismissible per user and per announcement with dismissal persisted, and respect effective/expiry windows.	FR-F15-04	US-122	Vitest + <code>fastify.inject()</code>
TC-I-071	F16	The background monitor probes every enabled registry row on schedule, records status/latency/history, advances the circuit state machine, and writes an issue row on each state transition — without blocking or being blocked by a user request.	FR-F16-01, FR-F16-02, FR-F16-03	US-125	<code>resilience.spec.ts</code>
TC-I-072	F16	<code>POST /api/admin/failure-injection</code> is administrator-only, audited, applies <code>UNAVAILABLE/SLOW/ERROR</code> per spoke, and auto-expires.	FR-F16-11	US-131	<code>resilience.spec.ts</code>
TC-I-073	F17	Seed volume is calibrated so every filter facet, for every role, returns at least one result on the default date range, and pagination is exercised on the investigator queue, the audit viewer, and the admin inventory.	FR-F17-01, FR-F17-07	US-133	Seed validation + integration tier
TC-I-074	F17	Persona binding: PER-01 Marcus Vale, PER-02 Dana Okonkwo, PER-03 Renée Ashford (SUBJ-00622), PER-04 Priya Raghunathan seed with their declared roles and attributes, including the multi-role identity.	FR-F17-02	—	Seed validation (FR-F17-02 AC-3)
TC-I-075	F17	Flagship preconditions P1–P8 are all present at baseline, including <code>ISS-2207 OPEN</code> , <code>CASE-A-1042 outstandingIssueCount=1</code> , the PDT and IM relationships, the spare second open issue, and ~120 pre-seeded audit events.	FR-F17-03, FR-F17-10	US-137	Seed validation (FR-F17-10)
TC-I-076	F17	Edge states are reachable without breaking anything: overdue items, blocked cases, a zero-item applicant, an above-tier case, and the one intentional orphan reference.	FR-F17-06	US-134	Integration tier
TC-I-077	F17	Seed validation fails startup with a specific, actionable message when any flagship precondition is missing or any persona binding is broken.	FR-F17-10	US-137	Seed validator (startup gate)
TC-I-078	F18	Each service starts and stops independently; <code>docker compose stop ual-im</code> is a real outage and the hub starts even when every spoke is down.	FR-F18-02	US-143, US-079	<code>resilience.spec.ts</code> + deployment dry run
TC-I-079	F18	Pre-flight checks report prerequisite versions and port availability with actionable messages; a port clash is resolvable from <code>.env</code> without editing code.	FR-F18-03	US-139	Deployment dry run
TC-I-080	F18	SCR-37 lists every running service and its state so a demo operator can confirm readiness at a glance.	FR-F18-07	US-142	Vitest + <code>fastify.inject()</code>
TC-I-081	F18	Clean shutdown stops all eleven containers with no orphaned processes and no held ports.	FR-F18-08	US-144	Deployment dry run
TC-I-082	F18	The documented troubleshooting section covers the three most likely demo-day failures and each documented remedy actually works.	FR-F18-09	US-144	Deployment dry run
TC-I-083	F19	The audit coverage and immutability suite runs in CI and a new mutating endpoint without an audit assertion fails the build.	FR-F19-05	US-148	<code>audit.coverage.spec.ts</code>
TC-I-084	F19	The reviewer-readable summary maps every success metric to a named test with a current status and maps every FRD requirement ID to its verifying test(s), flagging any requirement with no test.	FR-F19-09	US-151	FR-F19-09 CI artifact
TC-I-085	F19	The whole suite produces identical results across three consecutive runs and no test depends on the current wall-clock date; flaky tests are quarantined, never retried into passing.	FR-F19-10	US-151	CI determinism gate

8.3 Adapter-contract / conformance — 12 cases

TC ID	F	Assertion	FRD source	Stories	Verified by
TC-AC-001	F1	Every spoke verifies the Ed25519 principal assertion (<code>aud === self, exp > now</code>) and rejects a forged or expired assertion with <code>401 PRINCIPAL_REJECTED</code> ; <code>/health</code> and <code>/describe</code> require no assertion.	FR-F01-02, FR-F09-07	US-010	@ual/conformance + @ual/spoke-kit tests
TC-AC-002	F8	Conformance assertion 1: operations 1–7 are present on every adapter, and optional operations are present iff declared in <code>describe()</code> .	FR-F08a-01, FR-F08a-08	—	@ual/conformance
TC-AC-003	F8	Conformance assertion 2: <code>describe()</code> validates; every native status is mapped; every <code>requiredPermission</code> exists in the role matrix; an application declaring fewer actions degrades its UI affordances rather than erroring.	FR-F08a-02, FR-F08a-07	US-072, US-097	@ual/conformance

TC-AC-004	F8	Conformance assertions 3–4: every returned <code>WorkItem</code> satisfies the normalized shape; a scoped list for subject A returns zero rows for subject B; a scoped get for a foreign resource returns <code>NOT_FOUND</code> / <code>FORBIDDEN</code> , never data.	FR-F08a-03 , FR-F02-04	—	@ual/conformance
TC-AC-005	F8	<code>performAction</code> , <code>getWorkItemSummary</code> , and <code>getActivityHistory</code> behave to contract, and <code>stateVersion</code> changes after mutation and is stable otherwise.	FR-F08a-04	—	@ual/conformance
TC-AC-006	F8	Deadline honouring: a call with a 100 ms deadline aborts within 150 ms; a transient read failure retries per policy; a <code>performAction</code> timeout is never retried.	FR-F08a-05	US-073	@ual/conformance
TC-AC-007	F8	Error taxonomy: induced connection refusal, timeout, 4xx, 5xx, and malformed response each produce the correct <code>AdapterError</code> class, hub status, code, and user-facing message.	FR-F08a-06 , FR-F10-03	US-082	@ual/conformance
TC-AC-008	F8	Circuit behaviour: <code>circuitFailureThreshold</code> consecutive failures open the circuit; data calls then fail fast without touching the spoke; health probing continues; a successful probe closes it.	FR-F08a-05 , FR-F16-03	US-073	@ual/conformance
TC-AC-009	F8	No cross-spoke access: outbound calls target only this application's base endpoint, and zero spoke-to-spoke traffic occurs across the whole test run.	FR-F08a-08 , FR-F09-01	US-076, US-080	@ual/conformance + <code>isolation.spec.ts</code>
TC-AC-010	F8	The conformance suite is runnable standalone against any endpoint (<code>./run.sh conformance --adapter=CVS --endpoint=...</code>), and assertions 1, 2, and 6 run live during registration so a non-conformant application cannot be registered.	FR-F08a-08 , FR-F19-03	US-074, US-147	@ual/conformance
TC-AC-011	F9	Common spoke requirements verified identically on all six services: assertion verification, mandatory <code>X-UAL-Scope</code> on reads, deadline abort, idempotency records, <code>stateVersion</code> recompute, own activity row per mutation, honest <code>/health</code> , <code>/describe</code> , injection control, correlation echo, synthetic markers, and closed error set.	FR-F09-07	—	@ual/conformance + @ual/spoke-kit tests
TC-AC-012	F19	The full conformance suite runs in CI against all six adapters (10 assertions × 6) plus the cross-adapter behavioural assertions.	FR-F19-03	US-147	@ual/conformance in CI

8.4 End-to-end (browser) — 24 cases

TC ID	F	Assertion	FRD source	Stories	Verified by
TC-E-001	F0	CAC/PIV: method selection on SCR-01, synthetic certificate picker on SCR-02, selection establishes a session, and one <code>AUTH_SUCCESS</code> record is written.	FR-F00-01 , FR-F00-02	US-001, US-002	auth E2E (Playwright)
TC-E-002	F0	ECA: a distinct external-CA path with its own synthetic identity set establishes a session — demonstrably a second identity provider, not one provider with a skin.	FR-F00-03	US-003	auth E2E (Playwright)
TC-E-003	F0	Generic MFA: username entry then the deterministic demo one-time code (<code>UAL_DEMO_OTP</code>) establishes a session on a phone-width viewport.	FR-F00-04	US-004	auth E2E (Playwright)
TC-E-004	F1	A scripted traversal touching all five spokes in one session produces exactly one authentication event in the audit log.	FR-F01-03 , FR-F07a-06	US-009	flagship.spec.ts:authEventCount
TC-E-005	F3	Breadcrumbs express cross-application context (<code>Work Queue > eApp Case A-1042 > Related PVQ Issue ISS-2207</code>) with source badges, and each segment retraces correctly.	FR-F03-06	US-029	flagship.spec.ts + crawl
TC-E-006	F3	Link-and-control integrity crawl: every navigation item and every route in the SCR inventory, for all four roles — HTTP 200 or a designed error screen, non-empty <code><main></code> , unique <code><title></code> , exactly one <code><h1></code> , and every interactive control has a handler, a destination, or a documented disabled reason.	FR-F03-02 , FR-F19-08	US-032, US-150	crawl.spec.ts
TC-E-007	F4	Signing in as each of the four roles produces a substantively different, fully populated dashboard — different widgets, counts, and navigation — with no empty or placeholder widget.	FR-F04-02 , FR-F04-03 , FR-F04-04 , FR-F04-05	US-033, US-034, US-035, US-036	Dashboard E2E (Playwright)
TC-E-008	F4	Every dashboard widget, count, and alert links to a real populated destination, and the correct next item to work is reachable in two clicks or fewer for investigator and adjudicator.	FR-F04-06	US-039	crawl.spec.ts + dashboard E2E
TC-E-009	F5	Returning from a work item restores the queue's prior filters, sort, and page exactly.	FR-F05-09 , FR-F06-12	US-047	flagship.spec.ts + queue E2E
TC-E-010	F6	Each role completes at least one real, persisted action on its own work-item type (PDT designation approval, IEP task response, IM assignment acknowledgement), and the change is visible in both the spoke's data and the audit trail.	FR-F06-09 , FR-F06-10 , FR-F06-11 , FR-F06-04	US-057	Role-action E2E (Playwright)
TC-E-011	F7	FLAGSHIP. Sign in once as PER-01 via CAC/PIV → work queue default view → <code>EAPP:CASE-A-1042</code> → read Section 13A → related-items panel → <code>PVQ:ISS-2207</code> → resolve with disposition and narrative → SCR-20 dual-system confirmation reporting <code>Resolved – Substantiated</code> and <code>No outstanding issues</code> .	FR-F07a-02 , FR-F07a-03 , FR-F07a-04 , FR-F07b-01 , FR-F07b-04 , FR-F07b-05	US-059, US-060, US-061, US-062, US-063, US-064, US-065, US-067	flagship.spec.ts
TC-E-012	F7	FLAGSHIP continuity. Exactly one authentication event; zero document requests to any spoke origin; zero login forms after sign-in; zero identifiers typed beyond the narrative text; the shell stays mounted across the traversal with no interstitial, tab, iframe, or redirect.	FR-F07a-06 , FR-F01-03	US-059, US-061, US-069	flagship.spec.ts continuity block
TC-E-013	F7	FLAGSHIP dual-system proof. <code>spokeGet('PVQ','/issues/ISS-2207')</code> returns <code>RESOLVED_SUBSTANTIATED</code> with disposition, resolver, and narrative; <code>spokeGet('EAPP','/cases/CASE-A-1042')</code> returns <code>outstandingIssueCount = 0</code> and <code>REVIEW_COMPLETE_PENDING_ADJUDICATION</code> — both with the hub out of the path.	FR-F07a-05 , FR-F07b-05 , FR-F09-08	US-067	flagship.spec.ts:spokeGet × 2
TC-E-014	F7	FLAGSHIP repeatability. The flagship suite runs three consecutive times with a reset between each and produces identical results; <code>ISS-2207</code> is <code>OPEN</code> at the start of every run.	FR-F19-01 , FR-F17-11	US-136, US-140, US-145	flagship × 3 with <code>reset</code>
TC-E-015	F8	Registry-driven behaviour: removing an application row removes it cleanly from navigation, queue fan-out, admin inventory, and health monitoring with no code change and no errors; disabling it at runtime does the same without a redeploy.	FR-F08b-01 , FR-F08b-02 , FR-F08b-03	US-070, US-071	registration.spec.ts + architecture conformance
TC-E-016	F12	An administrator registers CVS through the five-step SCR-28 wizard in under five minutes with zero code changes and zero restarts, and it appears immediately in inventory, health, role navigation, and the work queue.	FR-F12-01 , FR-F12-06	US-094, US-099	registration.spec.ts
TC-E-017	F12	Post-registration propagation reaches an already-signed-in investigator's session: CVS work items appear without sign-out, reload, or re-authentication (registry-version poll).	FR-F12-05	US-098	registration.spec.ts , two browser contexts

TC - E - 018	F17	<code>reset</code> restores all seven namespaces and the hub to pristine baseline mid-demo, and the flagship path is immediately re-runnable.	FR-F17-11	US-136, US-099	<code>flagship × 3 with reset</code>
TC - E - 019	F18	From a clean checkout on a clean machine, one documented command brings up the database, migrations, seed, six spokes, hub, and UI, and a reviewer reaches a signed-in dashboard in under ten minutes.	FR-F18-01	US-138	Fresh-machine dry run
TC - E - 020	F18	A reviewer with no prior exposure follows <code>docs/README.md</code> — prerequisites, start command, URL, the four demo personas, the reset command — and succeeds unaided.	FR-F18-04	US-144	Fresh-machine dry run
TC - E - 021	F18	The flagship demo script's numbered steps 1–13 are drivable verbatim, and the expected observable state at each step matches, including the independent spoke verification step.	FR-F18-05	US-140	<code>flagship.spec.ts</code> (mirrors the script)
TC - E - 022	F18	All five secondary demo scripts are drivable: degraded-system behaviour (Segment 2), sixth-application registration (3), role-level zero trust (4), resource-level zero trust via <code>curl</code> (5b), and audit-chain review (6).	FR-F18-06	US-141	Secondary demo E2E (Playwright) + <code>curl</code> fixtures
TC - E - 023	F19	The flagship end-to-end suite exists, gates the build, and fails on regression to any asserted property, including the partial-failure variant.	FR-F19-01	US-145	<code>flagship.spec.ts</code>
TC - E - 024	F19	The crawl additionally scans user-facing copy on every route for prohibited content — stack-trace patterns, exception class names, hostnames, IPs, ports, SQL fragments, raw HTTP reason phrases, and the prohibited authentication verbs — and asserts every <code>entitlements.navigation[].href</code> resolves.	FR-F19-08	US-150	<code>crawl.spec.ts</code>

8.5 Accessibility — 20 cases

TC ID	F	Assertion	FRD source	Stories	Verified by
TC - A - 001	F0	Every authentication screen states that authentication is simulated, and the prohibited verbs ('verified', 'validated', 'authenticated against', 'trusted certificate') appear nowhere in user-facing copy.	FR-F00-08	US-008	<code>crawl.spec.ts</code> copy scan
TC - A - 002	F3	The shell provides landmark regions, a correct single- <code><h1></code> heading hierarchy, current-page indication, and a keyboard-operable primary navigation on every route.	FR-F03-01, FR-F14-05	US-026	<code>axe.spec.ts</code> + keyboard smoke
TC - A - 003	F3	The demo banner's verbatim copy is present in the accessible DOM on every route at 320px, 768px, and 1280px — including with a modal open and an <code>EMERGENCY</code> announcement active — with no close control and no <code>display:none</code> , <code>visibility:hidden</code> , or zero-height computed style; the <code>≤ units(15)</code> chrome budget holds at 320×568 and SCR-11's status sentence stays above the fold.	FR-F03-03, FR-F15-05, FR-F19-08	US-027, US-123	<code>crawl.spec.ts</code> banner assertion
TC - A - 004	F5	The work queue is operable with a screen reader: caption, header scope, sortable headers announcing sort state, and announced result counts on filter, search, and pagination change.	FR-F14-04	US-110	<code>axe.spec.ts</code> + screen-reader manual pass
TC - A - 005	F7	FLAGSHIP keyboard-only. The whole workflow is driven with Tab/Shift-Tab/Enter/Space/arrows/Escape; focus lands on the new page heading and the <code><title></code> changes descriptively at every screen boundary; no trap; axe reports zero serious or critical violations on SCR-13, SCR-15, SCR-16, and SCR-20.	FR-F14-02, FR-F07a-06, FR-F19-07	US-069	<code>axe.spec.ts</code> + keyboard smoke + recorded manual pass
TC - A - 006	F11	Administrator console tables follow the same accessible data-table pattern as F5 — caption, header scope, sortable headers, accessible pagination — so the console is as accessible as the rest of the product.	FR-F11-07, FR-F14-04	US-092	<code>axe.spec.ts</code>
TC - A - 007	F14	Every interactive element on every route is reachable and operable by keyboard alone; tab order matches DOM order; modals trap focus, close on Escape, and restore focus to the invoking control; the skip link is present, first focusable, and moves focus to <code><main></code> .	FR-F14-02, FR-F14-05, FR-F19-07	US-109, US-112	Keyboard smoke test (Playwright)
TC - A - 008	F14	Controls carry correct roles and accessible names; source attribution is in the accessible name, not a colour badge alone; asynchronous updates announce through live regions.	FR-F14-04, FR-F14-07	US-110	<code>axe.spec.ts</code> + manual screen-reader pass
TC - A - 009	F14	A visible focus indicator with sufficient contrast is present on every focusable control against its own background.	FR-F14-02, FR-F14-06	US-111	<code>axe.spec.ts</code> + keyboard smoke
TC - A - 010	F14	Forms: programmatically associated labels, described-by hint text, non-colour required indication, inline errors, and an error summary at the top with focus management and in-page links to the offending fields; submitting the resolution form empty is scanned in its error state.	FR-F14-03	US-113	<code>axe.spec.ts</code> (forms-in-error state)
TC - A - 011	F14	All text and meaningful non-text elements meet WCAG 2.1 AA contrast, and status is never conveyed by colour alone — always paired with text and/or icon, distinguishable in greyscale.	FR-F14-06	US-114	Automated contrast check + greyscale review
TC - A - 012	F14	Live regions announce queue refresh, action results, and degraded-system warnings without stealing focus; the dual-system confirmation deliberately does take focus.	FR-F14-07	US-115	<code>axe.spec.ts</code> + manual screen-reader pass
TC - A - 013	F14	Usable at 200% zoom and at 320px viewport width with no horizontal scrolling and no loss of function; <code>prefers-reduced-motion</code> is respected.	FR-F14-08	US-116	Responsive/zoom audit + <code>axe.spec.ts</code>
TC - A - 014	F14	Timing: the session warning is announced accessibly, extension is keyboard-operable, and no entered data is lost to the timer.	FR-F14-09, FR-F00-06	US-117	<code>axe.spec.ts</code> (open-modal state) + keyboard smoke
TC - A - 015	F14	Error, empty, and loading states are themselves accessible: SCR-30/31/32 scanned, empty states announced, and busy states exposed via <code>aria-busy</code> /live region rather than silence.	FR-F14-10	—	<code>axe.spec.ts</code> (non-default states)
TC - A - 016	F14	SCR-36 states the conformance target (Section 508 / WCAG 2.1 AA), known limitations, and the Q-01 style-guide assumption, and is reachable by anonymous users.	FR-F14-11, FR-F14-01	US-118	<code>axe.spec.ts</code> + <code>crawl.spec.ts</code>
TC - A - 017	F14	The gate. axe-core runs against every route × every role × five states (default, error, empty, degraded, loading, open modal, form-in-error) and the build fails on any serious or critical violation; moderate and minor are reported with owners and due dates.	FR-F14-12, FR-F19-06	US-149	<code>axe.spec.ts</code>
TC - A - 018	F15	No announcement, at any severity and with any modal open, obscures or displaces the demo banner.	FR-F15-05	US-123	<code>crawl.spec.ts</code> banner assertion
TC - A - 019	F19	The accessibility scan runs on every pull request, not only on main, and accessibility review is a merge gate.	FR-F19-06	US-149	CI configuration check

TC - A - 020	F19	The keyboard smoke suite runs per role across every route in CI and reports zero unreachable primary controls and zero traps.	FR-F19-07	US-149	Keyboard smoke test in CI
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8.6 RBAC negative-path — 11 cases

TC ID	F	Assertion	FRD source	Stories	Verified by
TC-RN-001	F2	An authenticated applicant requesting another subject's work item by ID is denied server-side, and the response is byte-identical (apart from <code>correlationId</code>) to the response for a fabricated ID; the denial is audited with its <code>policyRuleId</code> .	FR-F02-04 , FR-F02-07 , FR-F19-02	US-018	rbac.negative.spec.ts
TC-RN-002	F2	An investigator posting an adjudicator-only action — and any action absent from the server-computed list — receives 403; a stale <code>stateVersion</code> receives 409.	FR-F02-02 , FR-F02-05	US-019	rbac.negative.spec.ts
TC-RN-003	F2	Investigator, adjudicator, and applicant calls to every administrator endpoint and console route are refused server-side and audited.	FR-F02-02 , FR-F02-07 , FR-F11-07	US-020	rbac.matrix.spec.ts
TC-RN-004	F2	An investigator requesting a case outside their unit or assigned region is refused with <code>ATTR-INV-01</code> recorded in the audit record; the same case is absent from their queue fan-out.	FR-F02-03 , FR-F05-03	US-021	rbac.negative.spec.ts
TC-RN-005	F2	A <code>T3</code> investigator requesting a <code>T5</code> case is refused with <code>ATTR-INV-03</code> recorded in the audit record.	FR-F02-03 , FR-F17-06	US-022	rbac.negative.spec.ts
TC-RN-006	F2	An administrator requesting any mission work item is refused — administrators operate the platform, not the caseload — and no administrator path can mutate audit.	FR-F02-02 , FR-F11-07 , FR-F13-03	US-023	rbac.negative.spec.ts
TC-RN-007	F2	Any request supplying <code>role</code> , <code>activeRole</code> , or <code>principalId</code> in body or query is rejected with 400 by the reserved-field hook and has zero privilege effect.	FR-F10-01 , FR-F19-02	US-010, US-081	rbac.negative.spec.ts
TC-RN-008	F2	A mutating request without a CSRF token is refused with 403 and audited; an edited session cookie yields 401 and never an escalation.	FR-F10-01 , FR-F19-02	US-081	rbac.negative.spec.ts
TC-RN-009	F10	Coverage: 100% of endpoints in the route table have at least one unauthorized-access negative test; a new endpoint without one fails the build.	FR-F19-02 , FR-F10-02	US-085, US-146	rbac.negative.spec.ts (route-table generated)
TC-RN-010	F13	A mission user's audit query returns zero records authored by another actor; role-scoped visibility holds on viewer, chain view, and export.	FR-F13-07	US-107	rbac.negative.spec.ts
TC-RN-011	F19	The RBAC matrix suite is generated from the role matrix — every role × every action, permitted succeeding and non-permitted returning 403 with an <code>AUTHZ_DENIED</code> record — so adding a permission without a test is impossible.	FR-F19-02	US-146	rbac.matrix.spec.ts

8.7 Resilience / degradation — 15 cases

TC ID	F	Assertion	FRD source	Stories	Verified by
TC-R-001	F4	A slow or unavailable source degrades one dashboard widget, not the page; the widget names the affected application and what is missing, and clears automatically on recovery.	FR-F04-01 , FR-F16-05 , FR-F16-12	US-037, US-038	resilience.spec.ts
TC-R-002	F5	With IM forced offline, the queue renders the other four sources fully and actionably plus an explicit named, quantified notice ('Investigation Management is unavailable — 12 items are not shown'), and no error page appears anywhere.	FR-F05-05 , FR-F16-05	US-048	resilience.spec.ts
TC-R-003	F6	A work-item detail screen whose owning spoke is unavailable degrades cleanly: named warning, actions pre-disabled with reason, working exits — never a blank page.	FR-F06-01 , FR-F16-08 , FR-F16-05	US-058	resilience.spec.ts
TC-R-004	F7	FLAGSHIP partial-failure variant. With eApp injected <code>UNAVAILABLE</code> after the PVQ leg commits: <code>HTTP 207</code> , <code>overallOutcome = PARTIALLY_COMPLETED</code> , the word 'success' absent from response and rendered page, both per-system outcomes present, a working 'Retry eApp update' control, the PVQ disposition not reversed, automatic convergence after <code>cClearInjection</code> , and exactly one <code>ORCHESTRATION_INCOMPLETE</code> integration issue when retries exhaust.	FR-F07b-03 , FR-F11-03	US-066	flagship.spec.ts partial-failure variant
TC-R-005	F9	Stopping any single spoke process leaves the hub and the remaining five fully functional with a visible degraded warning; restarting restores it.	FR-F09-01 , FR-F18-02	US-079	resilience.spec.ts (one process-stop test per spoke)
TC-R-006	F15	An empty alerts panel means 'nothing needs you'; when the alert rules could not be computed, a distinct state says so rather than implying all is well.	FR-F15-01 , FR-F16-07	US-124	resilience.spec.ts
TC-R-007	F16	An action targeting an unavailable spoke is pre-emptively disabled with an explanatory message, and posting it anyway is refused server-side rather than failing mid-submission.	FR-F16-04	US-126	resilience.spec.ts
TC-R-008	F16	Every surface showing incomplete data carries a degraded warning naming the affected application and quantifying the gap.	FR-F16-05	US-127	resilience.spec.ts
TC-R-009	F16	With a <code>SLOW</code> injection, loading states render at widget and section granularity with accessible busy announcements and the page fills in rather than blanking.	FR-F16-06	US-128	resilience.spec.ts + axe.spec.ts (loading state)
TC-R-010	F16	A designed empty state and a degraded state are visually and non-visually distinguishable on every list, table, and widget.	FR-F16-07	US-129	resilience.spec.ts
TC-R-011	F16	An injected client-side exception renders SCR-32 inside the shell with working exits and a correlation ID — never a blank page and never a stack trace.	FR-F16-08 , FR-F16-10	US-130	resilience.spec.ts

TC-R-012	F16	Restoring a spoke clears the warning and restores data within 30 seconds with no reload and no re-authentication.	FR-F16-12	US-132	resilience.spec.ts
TC-R-013	F16	Fault-injection crawl: every route for every role with one spoke down — HTTP 200 or a designed error screen, non-empty <main>, demo banner present, zero uncaught client errors.	FR-F19-04	US-130, US-048	resilience.spec.ts + crawl.spec.ts
TC-R-014	F16	All sources down: the queue returns 200 with the all-unavailable empty state, never a 500.	FR-F19-04, FR-F16-07	US-129	resilience.spec.ts
TC-R-015	F19	The resilience suite covers all five single-spoke outage scenarios plus recovery per spoke, with zero error pages, and runs in CI using failure injection so suites stay parallelizable.	FR-F19-04	US-147	resilience.spec.ts

9. Non-Functional Requirement Traceability

Each NFR traced through three columns that matter separately: **where it is specified** as behaviour a builder can implement, **where it is implemented** architecturally, and **what verifies it**. An NFR with a specification but no verification is a promise; an NFR with verification but no specification is a test looking for a requirement. Both conditions are called out.

NFR	Category	Specified in (FRD)	Implemented in (TechArch)	Verified by (TC)	Metric
NFR-01	Accessibility — Section 508 / WCAG 2.1 AA, zero serious or critical	FR-F14-01 ... FR-F14-12, FR-F19-06	§12.4 accessibility architecture; §12.3 USWDS integration; §2.3 frontend stack	TC-A-017 (<i>the gate</i>), TC-A-019, TC-A-007, TC-A-005, TC-A-015 + documented manual keyboard and screen-reader pass	SM-07
NFR-02	Accessibility — never colour alone; keyboard-operable with visible focus	FR-F14-02, FR-F14-06	§12.4; §12.3 token-only theming	TC-A-009, TC-A-011, TC-A-007	SM-09
NFR-03	Design system — 100% USWDS v3, zero visual literals, tokens only	FR-F14-01, cited in FR-F08b, Y0a	§12.3; §2.3; <code>stylelint.config.cjs</code> at repo root; <code>packages/theme/_uswds-theme.scss</code> as the only file with a colour value — ADR-008	TC-U-006 (<i>Stylelint gate</i>), TC-A-016	—
NFR-04	Zero trust — every request authorised server-side at resource level	FR-F02-01, FR-F02-03, FR-F02-04, FR-F10-01, FR-F19-02	§8.3 PDP; §8.4 data-layer scoping; §1.3 choke point 1 — pipeline hook 7; §1.2 PDP + scoped spoke-query wrapper — ADR-016	TC-I-041 (<i>boot-time contract check</i>), TC-I-009, TC-RN-009 (<i>100% endpoint coverage</i>), TC-RN-011, TC-RN-001	SM-18
NFR-05	Zero trust — client-side hiding is presentation only	FR-F02-05, FR-F02-06	§8.5; §12.7 the frontend's security posture; §1.5 "the client renders what the server says it may see"	TC-I-008, TC-I-010, TC-RN-002, TC-RN-007	SM-18
NFR-06	Auditability — audit record written before the success response	FR-F13-01, FR-F10-05	§9.1 three layers of enforcement; §9.2 the interceptor at pipeline hook 9 (<code>onSend</code> , mutations)	TC-I-043 , TC-I-060, TC-I-026	SM-19
NFR-07	Auditability — append-only, sequence-numbered, hash-chained	FR-F13-03, FR-F19-05	§9.3 record structure and hash chain; §9.7 integrity verification; §3.8 final grants — <i>where immutability actually lives</i> — ADR-016	TC-I-063 (<i>routes + static analysis + DB grant + chain + sequence</i>), TC-I-083, TC-RN-006	—
NFR-08	Data integrity — no spoke reads another spoke's store	FR-F09-01, FR-F08a-01, FR-F17-04, Y0a, Y0b	§3.1 schema-per-service, role-per-service; §4.1, §4.2, §4.9 isolation verification checklist — ADR-004	TC-I-038 (<i>42 cross-schema probes raise 42501</i>), TC-U-005, TC-AC-009	SM-13
NFR-09	Resilience — single-spoke failure never blanks, throws, or takes the layer down	FR-F05-05, FR-F16-08, FR-F16-10, FR-F19-04	§10.3 partial-failure aggregation; §10.4 degraded-mode UX contract; §12.5 uniform state pattern — ADR-007	TC-R-002, TC-R-011, TC-R-013 (<i>fault-injection crawl</i>), TC-R-014, TC-R-005	SM-15, SM-16
NFR-10	Resilience — degraded state always visible and specific: app named, gap quantified	FR-F16-05, FR-F04-01, FR-F05-05	§10.4	TC-R-008 , TC-R-002, TC-R-001, TC-R-003	SM-15
NFR-11	Extensibility — registering a sixth application is configuration only	FR-F08b-02, FR-F08b-03, FR-F12-01, FR-F12-05, Y3	§17.1–17.6 the complete onboarding path and where the cost actually is; §5.4 registry record; §3.5 registry DDL — ADR-001	TC-E-016 , TC-E-017 , TC-E-015, TC-I-057	SM-11, SM-12
NFR-12	Privacy — no real PII or data; obviously synthetic	FR-F17-08, FR-F17-09, Y3	§13.7 obviously synthetic content; §4.1 per-record markers	TC-U-009 , TC-I-073, TC-I-074	—
NFR-13	Privacy — non-dismissible demo banner, verbatim in the accessible DOM on every route	FR-F03-03, FR-F15-05, FR-F19-08	§8.9 the demo banner as an architectural invariant ; §1.5 shell layer; §12.6 rendering strategy — server-rendered so there is no client state path to hide it	TC-A-003 (<i>3 viewports × modal × EMERGENCY announcement, evaluating accessible text</i>), TC-A-018, TC-E-006	SM-10
NFR-14	Usability — every navigation item resolves; zero dead links or placeholders	FR-F03-02, FR-F19-08	§12.1 routing — one segment per <code>SCR-nn</code> , "invent a route not in the screen inventory" prohibited; §1.5	TC-E-006 (<i>crawl, 4 roles × 38 routes</i>), TC-E-024, TC-E-008	SM-05, SM-06
NFR-15	Observability — correlation ID propagated into audit and error logs	FR-F01-06, FR-F10-03	§1.6 cross-cutting concerns — pipeline hook 1; §9.5 correlated chains; §10.7 integration issue recording	TC-I-007 , TC-I-042, TC-I-034, TC-I-037	SM-20
NFR-16	Responsiveness — usable from 320px and at 200% zoom	FR-F03-03, FR-F14-08	§12.3; §12.4	TC-A-013 , TC-A-003	SM-25
NFR-17	Performance (demo-grade) — dashboard and queue under 2 s; a slow spoke bounded	FR-F04-01, FR-F07a-02, FR-F08a-05, FR-F17-01	§10.2 timeout, retry, circuit policy; §10.3 aggregation within budget	TC-AC-006 (<i>deadline honoured within 150 ms</i>), TC-R-009, TC-I-016	—
NFR-18	Deliverability — clean checkout to running seeded app via one documented command	FR-F18-01, FR-F18-03	§15.1 the single command; §15.2 <code>compose.yaml</code> ; §0.5 topology and port map — ADR-002	TC-E-019 , TC-I-079, TC-E-020, TC-I-081	SM-21
NFR-19	Maintainability — a new work-item type or action is adapter-level only	FR-F02-05, FR-F07b-07, FR-F08a-07, FR-F08b-01	§5.5 capability negotiation; §17.5 a seventh application with a different backend; §17.6 where the onboarding cost actually is; §11.6 the engine is generic	TC-AC-003, TC-U-004 , TC-I-036	—
NFR-20	Security posture — sessions expire with warning; logout terminates all context; failed auth discloses nothing	⚠ Not cited by ID in any downstream document — functionally specified by FR-F00-02, FR-F00-06, FR-F00-07, FR-F01-04	§8.1 three simulated IdPs; §8.2 session and SSO propagation	TC-I-001, TC-I-002, TC-I-003, TC-A-014	—

Summary. 19 of 20 NFRs are specified by ID in at least one FRD requirement, implemented in at least one named TechArch section, and verified by at least one test case. NFR-20 is the exception and is carried as RTM-GAP-06: its behaviour is fully specified and fully tested, but no downstream document cites the identifier, so the NFR would not appear in a mechanically generated coverage report.

9.1 The four NFR clusters the Innovation Call will be judged on

Cluster	NFRs	Requirements	Architecture	Verification	Live demonstration
Accessibility (Section 508 / WCAG 2.1 AA)	NFR-01 , NFR-02 , NFR-03 , NFR-16	FR-F14-01...12 , FR-F19-06 , FR-F19-07	§12.3, §12.4, ADR-008, ADR-014	TC-A-001...TC-A-020 (20 cases); build fails on any serious or critical violation	Keyboard-only flagship pass, Segment 1
Zero-trust authorisation	NFR-04 , NFR-05	FR-F02-01...08 , FR-F10-01 , FR-F19-02	§8.3–8.7, §1.2, §1.3, ADR-016	TC-RN-001...TC-RN-011 (11 cases) + TC-I-041; 100% of endpoints carry a negative test	Segment 4 (role-level), Segment 5b (resource-level, live curl)
Auditability	NFR-06 , NFR-07 , NFR-15	FR-F13-01...08 , FR-F10-05 , FR-F19-05	§9.1–9.8, §3.8 grants, ADR-016	TC-I-043, TC-I-060...TC-I-066, TC-I-083, TC-RN-010	Segment 6 (one correlation ID resolving a whole cross-system action)
Graceful degradation	NFR-09 , NFR-10	FR-F16-01...12 , FR-F05-05 , FR-F19-04	§10.1–10.8, ADR-007	TC-R-001...TC-R-015 (15 cases)	Segment 2 (induce, observe, restore — two windows side by side)
Extensibility	NFR-11 , NFR-19	FR-F08a-*, FR-F08b-*, FR-F12-01...08	§5, §17, ADR-001	TC-E-015, TC-E-016, TC-E-017, TC-AC-002...TC-AC-012	Segment 3 (CVS registered live, under five minutes, zero restarts)

10. Success-Metric Traceability

All 25 metrics are demo-evaluable. Each row names the target, where the metric is specified downstream, the verifying test case, and the journey or demo segment in which a reviewer observes it.

10.1 Primary metrics — the demo passes or fails on these

Metric	Target	Specified in	Verifying test	Observed in
SM-01	Flagship completes end to end in a single session	FR-F07a-*, FR-F07b-*, FR-F19-01	TC-E-011, TC-E-023	JRN-01.01, Segment 1
SM-02	Exactly zero re-authentications — one AUTH_SUCCESS for the session	FR-F01-03 , FR-F07a-06 , FR-F18-05 , FR-F19-01	TC-E-012, TC-E-004	JRN-01.01 stage 1, Segment 1
SM-03	Both eApp and PVQ return updated state through their own APIs	FR-F07a-05 , FR-F07b-05 , FR-F09-08 , FR-F18-05	TC-E-013, TC-I-039	JRN-01.01 stage 9–10, Segment 1
SM-04	Zero manual re-entry of subject, case, or issue identifiers	FR-F03-06 , FR-F07a-06 , FR-F18-05	TC-E-012	JRN-01.01, JRN-02.01, Segment 1
SM-05	100% of primary navigation items, all four roles, resolve to a real populated page	FR-F03-02 , FR-F19-08	TC-E-006, TC-E-024	Crawl artifact, R2/R3 gates
SM-06	Zero dead links, placeholder screens, or buttons that do nothing	FR-F03-02 , FR-F19-08	TC-E-006, TC-E-008, TC-E-024	Crawl artifact + manual walkthrough

10.2 Accessibility and compliance

Metric	Target	Specified in	Verifying test	Observed in
SM-07	Zero serious or critical violations, all routes, all roles	FR-F14-12 , FR-F19-06 , FR-F07a-06	TC-A-017, TC-A-019, TC-A-005	CI accessibility report
SM-08	Flagship completable keyboard-only	FR-F14-02 , FR-F19-07	TC-A-005, TC-A-007	Recorded manual keyboard pass
SM-09	100% contrast conformance; status never colour alone	FR-F14-06	TC-A-011	Automated contrast check + greyscale review
SM-10	Banner present on 100% of routes incl. login and error pages; zero dismissal paths	FR-F03-03 , FR-F15-05 , FR-F19-08	TC-A-003, TC-A-018	Per-route crawl assertion

10.3 Architecture and extensibility

Metric	Target	Specified in	Verifying test	Observed in
SM-11	Sixth application registered live in under 5 minutes, zero code changes, zero restarts	FR-F08b-02 , FR-F12-01 , FR-F12-06 , FR-F18-06 , Y3	TC-E-016	JRN-04.01, Segment 3a
SM-12	Newly registered app appears in inventory, health, navigation, and queue immediately — including in an open session	FR-F03-02 , FR-F04-*, FR-F05-*, FR-F11-01 , FR-F12-05	TC-E-017	JRN-04.01 → JRN-01.02, Segment 3
SM-13	Zero cross-namespace data access; each spoke independently startable and queryable	FR-F08a-08 , FR-F09-01 , FR-F09-08 , FR-F19-03 , Y0b , Y1b	TC-I-038 (42 probes), TC-AC-009, TC-I-039	isolation.spec.ts + live API calls
SM-14	Investigator queue carries correctly attributed items from ≥4 of 5 spokes	FR-F04-*, FR-F05-02 , FR-F05-07	TC-I-017, TC-U-003	JRN-01.02 stage 1, R2 gate

10.4 Resilience

Metric	Target	Specified in	Verifying test	Observed in
SM-15	Forced outage produces a visible, specific degraded warning — never an error page or blank screen	FR-F05-05 , FR-F09-01 , FR-F16-05 , FR-F18-06 , FR-F19-04	TC-R-002, TC-R-008, TC-R-013	JRN-01.03, JRN-04.03, Segment 2a/2b
SM-16	With one spoke down, remaining sources render fully and remain actionable	FR-F05-05 , FR-F16-05 , FR-F19-04	TC-R-002, TC-R-007	JRN-01.03, Segment 2
SM-17	Restored spoke clears the warning and restores data without reload or re-authentication	FR-F16-12 , FR-F08a-05 , FR-F11-*, FR-F19-04	TC-R-012, TC-R-001	JRN-01.03 stage 7, JRN-04.03 stage 6

10.5 Security and audit

Metric	Target	Specified in	Verifying test	Observed in
SM-18	100% of endpoints authorise server-side; 100% of direct unauthorised calls denied	FR-F02-07 , FR-F10-01 , FR-F18-06 , FR-F19-02	TC-RN-009, TC-RN-011, TC-RN-001...008	Segment 4, Segment 5b (live <code>curl</code>)
SM-19	100% of state-changing actions produce exactly one audit record	FR-F10-05 , FR-F13-01 , FR-F19-05	TC-I-043, TC-I-060	<code>audit.coverage.spec.ts</code>
SM-20	Whole flagship workflow retrievable as one correlated chain	FR-F01-06 , FR-F07b-06 , FR-F13-04 , FR-F18-05 , Y3	TC-I-034, TC-I-007	SCR-34 audit chain view, Segment 1 and 6

10.6 Operability

Metric	Target	Specified in	Verifying test	Observed in
SM-21	Under 10 minutes from clean checkout to signed-in dashboard	FR-F18-01 , FR-F18-04	TC-E-019, TC-E-020	Fresh-machine dry run
SM-22	Flagship runnable 3 consecutive times with a reset between, identical result	FR-F05-*, FR-F07a-*, FR-F12-*, FR-F17-05 , FR-F17-11 , FR-F18-*, FR-F19-01	TC-E-014, TC-E-018, TC-U-008	Pre-demo rehearsal

10.7 Role experience

Metric	Target	Specified in	Verifying test	Observed in
SM-23	Four roles produce visibly and substantively different dashboards and nav sets — not a relabelled copy	FR-F02-*, FR-F04-02...05	TC-E-007, TC-I-013	Four sign-ins side by side
SM-24	Correct next item reachable from the dashboard in ≤ 2 clicks, investigator and adjudicator	FR-F04-06	TC-E-008, TC-I-068	JRN-01.02, JRN-02.02, demo script observation
SM-25	Applicant answers "where am I" within 30 s at 320×568 without scrolling, zero unexplained internal terms	FR-F03-03 , FR-F04-04 , FR-F19-08	TC-A-003 (chrome budget + above-the-fold assertion), TC-A-013, TC-E-007	JRN-03.01, Segment 5a, on a phone-width viewport

Summary. 25 of 25 metrics have a named verifying test case and a named observation point. Two reporting inconsistencies are carried as gaps: FR-F19-09 scopes its CI traceability artifact to "SM-01 through SM-22" while the PRD F19 acceptance signal requires SM-01 through SM-25 (RTM-GAP-07), and the UserStories success-metric coverage table also stops at SM-22 (RTM-GAP-08).

11. Coverage Gap Analysis

No blocking gaps. Every PRD feature has FRD requirements, every journey has covering stories, every story has a test case, and every requirement has a test case. The findings below are honest disclosures and, in four cases, reporting defects worth fixing before hand-off. Each carries an ID so it can be closed rather than rediscovered.

11.1 Mandatory checks — results

Check	Method	Result
Any PRD feature with no FRD requirement ?	Grouped 182 FR-IDs by originating feature	None. 20 of 20 features covered, min 6, max 12 requirements.
Any FRD requirement with no user story ?	Parsed **FRD:** trailers on all 151 stories, reversed the mapping	15 of 182 — see RTM-GAP-01 ... RTM-GAP-04 . All 15 are machine-verified rather than user-observable; all 15 have a test case.
Any user story with no test case ?	Reversed the 176-case TC catalogue against the 151 story IDs	None. 151 of 151 covered.
Any FRD requirement with no test case ?	Reversed the TC catalogue against the 182 FR-IDs	None. 182 of 182 covered.
Any journey with no covering stories ?	STORY-MAP journey coverage + per-release "closed by" attributions	None. 10 of 10 journeys, 78 of 78 stages covered.
Any JTBD with no journey ?	JOURNEYS JTBD coverage check	None. 16 of 16 jobs exercised.
Any persona with < 2 journeys ?	JOURNEYS journey index	None. Each of PER-01...04 carries 2–3 journeys.
Any dangling FR- reference anywhere in the suite?	Set comparison across FRD, TechArch (56), UserStories (168), JOURNEYS (4), STORY-MAP (2), UX-Mockup (64)	None. Zero dangling across 294 unique downstream citations.
Any story not placed on the story map?	STORY-MAP placement integrity	None. 151 placed, 0 orphans.
Any SCR-nn reference to a screen outside the inventory?	Set comparison against FR-F03-02	None. SCR-01...38 closed.

11.2 Requirements with no citing user story — RTM-GAP-01 ... RTM-GAP-04

Fifteen requirements are not cited by any story's ****FRD:**** trailer. In every case the behaviour is either a machine-verified contract, an internal lifecycle detail, or a documentation artifact — nothing user-observable is missing. All fifteen are nevertheless test-covered, and the implicit story is named so the link can be made explicit in a future revision.

Gap	Requirements	Why no story exists	Test coverage	Implicit story	Recommendation
RTM-GAP-01 (session internals)	FR-F01-01 single server-side session issuance; FR-F01-04 per-spoke context handles and invalidation	Both are internals of behaviour a user experiences through other stories: the session exists (US-009, US-012) and ends (US-007)	TC-U-002, TC-I-003	US-009 / US-012 for issuance, US-007 for invalidation	Add the FR citations to US-009 and US-007. No new story needed.

RTM-GAP-02 <i>(contract and genericity)</i>	FR-F07b-07 orchestration is not special-cased; FR-F08a-01 interface operations; FR-F08a-03 <code>listWorkItems / getWorkItem</code> ; FR-F08a-04 <code>performAction / getWorkItemSummary / getActivityHistory</code> ; FR-F08a-06 adapter error taxonomy; FR-F08b-04 role visibility and per-application access scoping; FR-F09-07 common spoke service requirements	These are the adapter seam itself — verified by conformance suite, not by a user action. A user story for "the interface has seven operations" would be a fiction.	TC-U-004, TC-AC-002, TC-AC-004, TC-AC-005, TC-AC-007, TC-I-036, TC-AC-011	US-070 and US-074 <i>(prove a new adapter is correct before trusting it)</i> ; US-082 for the error taxonomy	Cite these FRs from US-074 and US-082. FR-F07b-07 is on the flagship thread and should be cited from US-064 explicitly, because "the engine is generic" is the claim that makes F12 credible.
RTM-GAP-03 <i>(API conventions)</i>	FR-F10-02 endpoint groups and their screens; FR-F10-04 pagination/filtering/sorting conventions; FR-F10-07 rate limiting (demo-grade)	Cross-cutting conventions rather than features. FR-F10-07 is the only requirement in the suite with neither a story nor an upstream PRD capability bullet — it is a defensive addition made in the FRD.	TC-I-045, TC-I-046, TC-I-047	US-045 <i>(pagination)</i> , US-084 <i>(API documentation)</i> ; none for rate limiting	Cite FR-F10-04 from US-045 and FR-F10-02 from US-084. Decide explicitly whether FR-F10-07 is in scope for the prototype or should be marked [WITHDRAWN] .
RTM-GAP-04 <i>(cross-cutting and documentation)</i>	FR-F12-08 onboarding documentation; FR-F14-10 error/empty/loading state accessibility; FR-F17-02 personas and role/attribute coverage	FR-F14-10 is the notable one: it is a P0 accessibility requirement on exactly the states most likely to be built carelessly, and it has no story of its own. FR-F12-08 and FR-F17-02 are artifacts whose consumers are other documents.	TC-I-059, TC-A-015 , TC-I-074	US-128, US-129, US-130 collectively cover the states; US-118 covers the accessibility claim; US-133 covers persona seeding	Add explicit FR-F14-10 citations to US-128, US-129, and US-130. A P0 accessibility requirement reachable only by inference is the kind of gap that produces a demo-day violation.

11.3 Reporting and direction gaps — RTM-GAP-05 ... RTM-GAP-11

Gap	Finding	Impact	Recommendation
RTM-GAP-05	JOURNEYS-DCSA-UAL.md contains zero US- references across all 13 chunks. Journey → story traceability exists only in STORY-MAP .	Medium. Coverage is verifiable, but only by joining two documents. A journey stage that loses its story would not be detectable from JOURNEYS alone.	Add a story column to each journey's stage table, or a per-journey traceability block mirroring §6.3 of this document.
RTM-GAP-06	NFR-20 (security posture — session expiry, logout termination, non-disclosing auth failure) appears only in the PRD . No FRD requirement, TechArch section, or story cites the identifier.	Low functionally, medium for reporting. The behaviour is fully specified (FR-F00-02 , FR-F00-06 , FR-F00-07 , FR-F01-04) and fully tested (TC-I-001, TC-I-002, TC-I-003, TC-A-014), but a mechanically generated NFR coverage report would show it uncovered.	Add NFR-20 to the <code>**Traces to:**</code> line of FRD/F00-simulated-mfa-authentication.md and F01-unified-session-sso.md .
RTM-GAP-07	FR-F19-09 rule 1 scopes the CI traceability artifact to "SM-01 through SM-22 ". The PRD F19 acceptance signal requires "every success metric SM-01 through SM-25 ". SM-23 and SM-24 are cited by no F19 requirement at all.	Medium. Three role-experience metrics — including SM-24 , a two-click usability claim the demo script depends on — would be absent from the reviewer-readable report.	Correct FR-F19-09 rule 1 to SM-25 and add SM-23 / SM-24 assertions to FR-F19-08 (crawl) and the dashboard E2E. TC-E-007 and TC-E-008 already carry them.
RTM-GAP-08	The UserStories success-metric coverage table stops at SM-22 ; SM-23 , SM-24 , SM-25 have no story-level demonstration named.	Low. FRD F04 covers all three; the omission is in the story document's summary table only.	Extend the table: SM-23 → US-033...036; SM-24 → US-039; SM-25 → US-035, US-116.
RTM-GAP-09	TechArch cites 56 of 182 FR-IDs explicitly by ID; the remaining 126 are covered structurally by section but carry no ID-level citation.	Low. The chunk map (§0.7) and the per-feature mapping in §3 of this document make coverage determinable. But there is no mechanical way to prove TechArch covers a given requirement.	Add a <code>**Implements:**</code> line to each TechArch section listing the FR-IDs it realises. This is the single highest-leverage traceability improvement available to the suite.
RTM-GAP-10	No TC- identifier space existed before this RTM. FR-F19-09 rule 5 requires the CI artifact to map every FRD requirement to its verifying test(s).	Medium. Without an adopted TC space, the CI artifact will invent its own identifiers and the two traceability views will diverge.	Adopt the 176 TC IDs in §8 as the build's test-registry keys, and have FR-F19-09's artifact emit them. Then this RTM and the CI report are the same claim in two formats.
RTM-GAP-11	The reconciliation hand-off reported 186 defined / 186 referenced FR-IDs. Re-extraction gives 182 / 182 . Integrity is exact (zero dangling, zero orphans); the count was stale.	Low. Corrected here.	Update the FRD's own summary count to 182 wherever it appears, so the next reconciliation does not re-open a closed question.

11.4 Disclosed scope and sequencing risks — RTM-GAP-12 ... RTM-GAP-14

Gap	Finding	Consequence to manage
RTM-GAP-12	Q-01 (DCSA Ecosystem Style Guide, Attachment 1) and Q-02 (Page_render_reference.pdf) remain open — external input required ; neither can be closed from inside this effort.	Blast radius is bounded and documented: A-05 in FRD Y3 confines it to F14 theming tokens and header assets — a token and asset swap, not a component rewrite. Disclosed on SCR-36 per FR-F14-11 and verified by TC-A-016. Revisit immediately on receipt of Attachment 1 and record any divergence as a change to F14.
RTM-GAP-13	JRN-01.02 stage 7 and JRN-02.02 stages 5–6 are stranded until R3 — both journeys are mostly demonstrable from R1/R2 but cannot be walked end to end earlier.	Deliberate: stage 7 is the <i>consequence</i> of registration (F12) and stages 5–6 are the <i>consequence</i> of the health monitor (F16). Rehearsal scripts must not claim these journeys complete before R3 . Closed by US-098 and US-129 respectively.
RTM-GAP-14	PER-02 and PER-03 can do no meaningful work at the end of R1 — a deliberate consequence of the walking-skeleton rule.	If the build stops after R1, the RBAC/ABAC claim is unproven , because its clearest live demonstration (US-017 — the same work item, two roles, two server-computed action sets) requires PER-02 to exist. R2 is therefore not optional for a credible evaluation. Additionally, all of PER-04's journeys land in R3, which carries 50 stories and 5 of the 10 journeys; mitigated by shipping US-070 (registry-driven fan-out) and US-145 (flagship regression test) in R1 and by R3's internal ordering gate sequencing operability before extensibility.

11.5 Priority of remediation

Priority	Gaps	Rationale
Fix before build starts	RTM-GAP-04 <i>(FR-F14-10 citations)</i> , RTM-GAP-10 <i>(adopt the TC space)</i>	A P0 accessibility requirement reachable only by inference, and a test registry that will otherwise fork.
Fix before hand-off	RTM-GAP-06 , RTM-GAP-07 , RTM-GAP-08 , RTM-GAP-11	All four are one-line reporting corrections that make coverage mechanically provable.
Fix opportunistically	RTM-GAP-01 , RTM-GAP-02 , RTM-GAP-03 , RTM-GAP-05 , RTM-GAP-09	Traceability quality improvements; no behaviour is missing.
Monitor, do not fix	RTM-GAP-12 , RTM-GAP-13 , RTM-GAP-14	Bounded external dependency and two deliberate sequencing decisions.

12. Change Management

12.1 Change control for this matrix

This RTM is a **derived document**. It is regenerated rather than edited whenever any traced document changes, and the regeneration re-extracts every identifier rather than trusting the previous version. The following changes require regeneration:

Trigger	What must be re-verified
A PRD feature is added, removed, or re-prioritised	§2.4, §3, §4; the feature's FR set, story set, and TC set
An FRD requirement is added, withdrawn, or renumbered	§5 in full; §11.1 requirement-coverage checks; the affected feature's TC assignments
A user story is added, split, or removed	§5 story columns; §6.2–6.4; §11.1 story-coverage check; STORY-MAP placement integrity
A TechArch section, ADR, table, or endpoint changes	§3 TechArch column; §9 implementation column; §7.4 if the flagship thread is touched
An NFR or SM is added or retargeted	§9 or §10 in full
A persona, job, journey, or journey stage changes	§6 in full; §11.1 human-layer coverage checks
Attachment 1 (DCSA Ecosystem Style Guide) is received	RTM-GAP-12 closes; F14 theming requirements and TC-A-016 re-verified; PRD Q-01 re-dispositioned

Rule. A requirement is never renumbered. A retired requirement is marked [WITHDRAWN] in place, and this matrix retains its row with the withdrawal recorded — so a build artifact referencing an old ID resolves to an explanation rather than to nothing.

12.2 Change log

Version	Date	Author	Change	Affected sections
1.0	2026-09-15	Pivota Spec Framework — RTM generator	Initial matrix. Traced 20 PRD features, 182 FRD requirements, 18 TechArch chunks / 16 ADRs / 58 tables / 34 endpoints, 151 user stories, 4 personas, 16 jobs, 10 journeys, 3 releases, 20 NFRs, 25 success metrics. Introduced the 176-case TC- identifier space. Recorded 14 traceability gaps. Corrected the FR-ID count from the reported 186 to the verified 182 with integrity confirmed.	All

12.3 Traceability health indicators

Track these five numbers across revisions. A regression in any of them is a traceability defect, not a documentation preference.

Indicator	Baseline v1.0	Target
FRD requirements with no citing story	15 / 182 (8.2%)	≤ 5 / 182, and zero among P0 user-observable requirements
FRD requirements with no explicit TechArch ID citation	126 / 182 (69.2%)	0 — via the **Implements:** line proposed in RTM-GAP-09
FRD requirements with no test case	0 / 182	0
User stories with no test case	0 / 151	0
NFRs not cited by ID downstream	1 / 20 (NFR-20)	0
Success metrics absent from the CI traceability artifact	3 / 25 (SM-23 , SM-24 , SM-25)	0
Dangling requirement references, any document	0	0

13. Approval and Sign-Off

By signing below, each approver confirms that the traceability relationships in this matrix accurately reflect the documents named in §12.1 as of the stated date, that the gaps recorded in §11 have been reviewed and dispositioned, and that the remediation priorities in §11.5 are accepted.

Role	Responsibility in this matrix	Name	Signature	Date
Product Owner	PRD features, priorities, product principles, success metrics; accepts §3, §4, §10	_____	_____	_____
Lead Business Analyst	FRD requirement completeness and the PRD→FRD mapping; accepts §5 and §11.2	_____	_____	_____
Lead Architect	TechArch specifications, ADRs, data model, adapter seam; accepts §3 TechArch column, §7.4, §9 implementation column	_____	_____	_____
UX / Accessibility Lead	Personas, jobs, journeys, screen inventory, Section 508 / WCAG 2.1 AA conformance; accepts §6, §9.1 accessibility cluster, and the RTM-GAP-04 remediation	_____	_____	_____
QA / Verification Lead	Test-case catalogue, coverage properties, CI gates; accepts §8, §10, and the RTM-GAP-10 adoption of the TC- space	_____	_____	_____
Security / Zero-Trust Reviewer	NFR-04 ... NFR-07 , RBAC negative-path coverage, audit immutability; accepts §9.1 zero-trust and auditability clusters	_____	_____	_____
Demo Operator	Demonstrability of the six scripted segments and the flagship thread; accepts §7 and §11.4 RTM-GAP-13	_____	_____	_____

13.1 Conditions of approval

1. **RTM-GAP-04 and RTM-GAP-10 are closed before build start.** A P0 accessibility requirement (FR-F14-10) must be cited by the stories that deliver it, and the TC- identifier space must be adopted as the build's test registry so that FR-F19-09 's artifact and this matrix remain the same claim.
2. **RTM-GAP-06, RTM-GAP-07, RTM-GAP-08, and RTM-GAP-11 are closed before hand-off.** Four one-line corrections that make NFR and success-metric coverage mechanically provable.
3. **The flagship thread in §7 is re-verified at every release gate.** R1, R2, and R3 each assert SM-01 ... SM-04 , SM-08 , SM-20 , and SM-22 ; a regression in TC-E-011 , TC-E-012 , TC-E-013 , or TC-R-004 blocks the gate regardless of what else passes.
4. **RTM-GAP-12 is revisited on receipt of Attachment 1.** The DCSA Ecosystem Style Guide assumption is disclosed on SCR-36 and must be re-dispositioned, not silently retained, once the real guide exists.
5. **Approval of this matrix is not approval of the specifications it traces.** It is a statement that the relationships between them are complete, consistent, and gap-disclosed.

14. Related Documents

Document	Relationship to this matrix
.planning/PROJECT.md	Source charter — the core-value statement that §7 traces, plus constraints and key decisions
project_specs/PRD-DCSA-UAL.md	Upstream — features F0 ... F19 , principles PRIN-01 ... PRIN-06 , NFR-01 ... NFR-20 , SM-01 ... SM-25 , risks R-01 ... R-15 , questions Q-01 ... Q-07
project_specs/FRD-DCSA-UAL.md (chunks in FRD/)	Upstream — 182 functional requirements, the 38-screen inventory (FR-F03-02), hub and spoke schemas (Y0a , Y0b), endpoint catalogues (Y1a , Y1b), error catalogue (Y2), integration points (Y3)
project_specs/TechArch-DCSA-UAL.md (chunks in TechArch/)	Upstream — 18 chunks, ADR-001 ... ADR-016 , component map, adapter seam, security placement, audit enforcement, resilience, orchestration, testing architecture
project_specs/UserStories-DCSA-UAL.md (chunks in UserStories/)	Upstream — 151 stories in 20 epics, story index, priority tiers
project_specs/PERSONAS-DCSA-UAL.md	Upstream — PER-01 ... PER-04 , feature-persona matrix
project_specs/JTBD-DCSA-UAL.md	Upstream — JTBD-01.1 ... JTBD-04.4 , outcome-to-feature traceability
project_specs/JOURNEYS-DCSA-UAL.md (chunks in JOURNEYS/)	Upstream — JRN-01.01 ... JRN-04.03 , 78 stages, the six-segment scripted demonstration path, journey → JTBD traceability
project_specs/STORY-MAP-DCSA-UAL.md (chunks in STORY-MAP/)	Upstream — backbone matrix, releases R1 / R2 / R3 , JTBD and journey coverage, placement integrity, gap analysis
project_specs/UX-Mockup-DCSA-UAL.md	Upstream — screen and flow designs; 64 FR-ID citations, all resolving
project_specs/ref_docs/DCSA_Innovation_Call_01_Unified_Application_Layer.pdf.md	Governing requirement (input, not specification)
project_specs/ref_docs/DCSA_Innovation_Gateway_20260623.pdf.md	Parent solicitation — technical priorities and evaluation criteria
Missing — "Attachment 1 — DCSA Ecosystem Style Guide"	Referenced by the Innovation Call, not supplied . Drives Q-01 and RTM-GAP-12 .